# JEFFERSON COUNTY
# BOETTCHER MANSION
# CHILLER REPLACEMENT

## GOLDEN, CO 80401




SITE LEVEL KEY MAP

## DRAWING PACKAGE

SHEET INDEX

DELIVERABLE KEY:
ISSUED '●'
DISCONTINUED 'X'
REFERENCE ONLY 'O'

| NUMBER | NAME | 90% CHECK SET - 09.11.2026 | 100% CONSTRUCTION DOCUMENTS - 09.22.2026 |
|---|---|---|---|
| G-000 | PROJECT COVER SHEET | ● | ● |
| | | | |
| M-000 | MECHANICAL LEGENDS & SCHEDULES | ● | ● |
| M-101 | PARTIAL BASEMENT MECHANICAL DEMOLITION PLAN | ● | ● |
| M-201 | PARTIAL BASEMENT MECHANICAL NEW WORK PLAN | ● | ● |
| M-301 | MECHANICAL ISOMETRIC - DEMOLITION | ● | ● |
| M-302 | MECHANICAL ISOMETRIC - NEW WORK | ● | ● |
| M-400 | MECHANICAL SCHEMATICS | ● | ● |
| M-500 | MECHANICAL DETAILS | ● | ● |
| M-600 | MECHANICAL CONTROLS | ● | ● |
| M-700 | MECHANICAL SCHEDULES | ● | ● |
| | | | |
| E-000 | ELECTRICAL LEGENDS | ● | ● |
| E-100 | PARTIAL BASEMENT ELECTRICAL DEMOLITION PLAN | ● | ● |
| E-200 | PARTIAL BASEMENT ELECTRICAL NEW WORK PLAN | ● | ● |
| E-500 | ELECTRICAL SCHEDULES & ONE-LINE DIAGRAM | ● | ● |

## PROJECT TEAM

### DESIGN TEAM

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### AUTHORITY HAVING JURISDICTION

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## PROJECT DESCRIPTION

### PROJECT LOCATION

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLOROW RD
GOLDEN, CO 80401

### PROJECT SCOPE OVERVIEW

REPLACE IN KIND THE EXISTING AIR-COOLED CHILLER AND CHILLED WATER SYSTEM PUMPS, INCLUDING ASSOCIATED APPURTENANCES AND PIPING BETWEEN THE CHILLER AND PUMPS. PROVIDE A NEW CHILLED WATER BUFFER TANK TO INCREASE SYSTEM VOLUME AND REDUCE CHILLER SHORT CYCLING DURING PERIODS OF LOW DEMAND. REPLACE THE EXISTING GLYCOL FEEDER SYSTEM, AIR SEPARATOR, EXPANSION TANK, AND ASSOCIATED SYSTEM COMPONENTS AS INDICATED.

### APPLICABLE CODES

| | |
|---|---|
| INTERNATIONAL BUILDING CODE (IBC) | 2024 |
| INTERNATIONAL EXISTING BUILDING CODE (IEBC) | 2024 |
| INTERNATIONAL FIRE CODE | 2024 |
| INTERNATIONAL MECHANICAL CODE | 2024 |
| INTERNATIONAL PLUMBING CODE | 2024 |
| INTERNATIONAL FUEL GAS CODE | 2024 |
| NATIONAL ELECTRICAL CODE (NEC / NFPA 70) | 2023 |
| INTERNATIONAL ENERGY CONSERVATION CODE | 2024 |

### BUILDING INFORMATION

| | |
|---|---|
| OCCUPANCY TYPE (PER IBC CHAPTER 3) | GROUP B |

## PROJECT GENERAL NOTES

GENERAL NOTES

1. CONFORM TO THE REQUIREMENTS OF ALL LOCAL, CITY, COUNTY, STATE AND FEDERAL AGENCIES WHICH HAVE AUTHORITY OVER THE PROJECT. INCLUDE ALL ITEMS OF LABOR AND MATERIAL REQUIRED TO MEET SUCH REQUIREMENTS REGARDLESS OF THE FAILURE TO SPECIFY IN THE PROJECT MANUAL OR INDICATED ON THE DRAWINGS EACH INDIVIDUAL ITEM.
2. VERIFY THE EXISTING CONDITIONS AT THE PROJECT SITE BEFORE SUBMITTING COST PROPOSAL. BE ADVISED THAT ALL LOCATIONS ARE APPROXIMATE.
3. DO NOT SCALE DRAWINGS. VERIFY DIMENSIONS PRIOR TO COMMENCEMENT OF WORK.
4. THESE DRAWINGS ARE NECESSARILY DIAGRAMMATIC IN NATURE. NOT ALL FITTINGS, OFFSETS, AND SLOPES ARE SHOWN. THE CONTRACTOR SHALL VISIT THE SITE PRIOR TO BIDDING AND INCLUDE ALL FITTINGS, OFFSETS, AND SLOPES AS REQUIRED TO PROVIDE A COMPLETE AND FUNCTIONING SYSTEM.
5. UNLESS SPECIFICALLY DESIGNATED OTHERWISE, THIS CONTRACTOR SHALL FURNISH AND/OR INSTALL EACH AND EVERY ITEM AND EQUIPMENT SHOWN OR DESCRIBED, TOGETHER WITH ALL COMPONENTS AND INCIDENTALS (MATERIAL LABOR, FEES, TRANSPORTATION, TOOLS, STORAGE, CUTTING, PATCHING, TRENCHING, CLEAN-UP, ETC.) NECESSARY TO COMPLETE THE WORK WITH NO ADDITIONAL EXPENSE TO OWNER.
6. IN CASE OF DIFFERENCES BETWEEN BUILDING CODES, SPECIFICATIONS, STATE LAWS, LOCAL ORDINANCES, INDUSTRY STANDARDS, UTILITY COMPANY REGULATIONS, AND THE CONTRACT DOCUMENTS, THE MOST STRINGENT SHALL GOVERN. SHOULD THE CONTRACTOR PERFORM ANY WORK THAT DOES NOT COMPLY WITH THE REQUIREMENTS OF THE APPLICABLE BUILDING CODES, SPECIFICATIONS, STATE LAWS, LOCAL ORDINANCES, INDUSTRY STANDARDS, AND ORDINANCES, INDUSTRY STANDARDS, AND UTILITY COMPANY REGULATIONS, THEY SHALL BEAR ALL COST ARISING IN CORRECTING THESE DEFICIENCIES.
7. LOCATIONS, QUANTITIES AND OTHER MEASUREMENTS OF THE BUILDING AND EQUIPMENT ARE OFFERED AS A GENERAL GUIDE ONLY, WITHOUT GUARANTEE AS TO COMPLETE ACCURACY. THE CONTRACTOR SHALL EXAMINE THE DRAWINGS AND EQUIPMENT AND VERIFY TO HIS OWN SATISFACTION; ALL CONDITIONS AND THEIR RELATIONS AND EFFECT REQUIRED FOR A COMPLETE AND CORRECTLY OPERATING MECHANICAL SYSTEM.
8. EACH CONTRACTOR SHALL VISIT THE SITE, ASCERTAIN THE CONDITIONS TO BE MET THEREIN IN INSTALLING THE WORK, AND MAKE DUE PROVISIONS IN BID. FAILURE TO COMPLY WITH THIS REQUIREMENT SHALL NOT BE CONSIDERED AS JUSTIFICATION FOR THE OMISSION OF WORK, FAULTY WORK OR FOR PAYMENT OF EXTRA COMPENSATION.
9. AT ALL TIMES, TAKE CARE TO PROTECT APPARATUS FROM DAMAGE. ADEQUATELY PROTECT STORED APPARATUS AT THE SITE. PROVIDE TARPAULINS OR OTHER PROTECTION COVERING AS REQUIRED TO THE SATISFACTION OF THE OWNER.
10. LOCATE ALL EQUIPMENT, FIXTURES, PIPING, ETC., APPROXIMATELY IN GENERAL LOCATION AS SHOWN, BUT CONFORM TO ALL STRUCTURAL AND FINISH CONDITIONS OF BUILDING.
11. COOPERATE WITH OTHER CONTRACTORS TO KEEP THE AREA CLEAN AND FREE OF UNNECESSARY MATERIALS OR DEBRIS. WHEN WORK IS COMPLETE, REMOVE ALL TOOLS, SCAFFOLDING, MACHINERY, AND DEBRIS. BROOM SWEEP THE AREA.
12. COOPERATE WITH THE OWNER AND GENERAL CONTRACTOR TO MEET CONSTRUCTION SCHEDULES. COOPERATE IN ESTABLISHING THESE SCHEDULES AND AVOID DELAYING OTHER CONTRACTORS.
13. PREPARE AS-BUILT DRAWINGS INDICATING ACTUAL ROUTINGS OF UTILITIES INCLUDING LABELING OF EACH LINE SHOWING SIZE, ETC.
14. AT TIME OF FINAL INSPECTION, TEST ALL CONNECTIONS AND OPERATION OF ALL EQUIPMENT. ALL EQUIPMENT AND SYSTEMS SHALL BE DEMONSTRATED TO OWNER & ENGINEER.
15. BRING TO THE ATTENTION OF THE ENGINEER ANY DISCREPANCIES SO THAT AN ADDENDUM MAY BE ISSUED BEFORE BIDDING. ANY OMISSIONS OR ERRORS FOUND AFTER BIDDING WILL BE AT THE EXPENSE OF THE CONTRACTOR.

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CHILLER REPLACEMENT
JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLOROW RD, GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME
PROJECT COVER SHEET

SHEET NO.
G-000


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## PIPE SYSTEM DESIGNATIONS

### HYDRONIC SYSTEMS

| | |
|---|---|
| CR | CONDENSER RETURN |
| CS | CONDENSER SUPPLY |
| CCR | CONDENSER RETURN - CLOSED |
| CCS | CONDENSER SUPPLY - CLOSED |
| CHR or CWR | CHILLED WATER RETURN |
| CHS or CWS | CHILLED WATER RETURN |
| RFR | RADIANT FLOOR RETURN |
| RFS | RADIANT FLOOR SUPPLY |
| GF | GLYCOL FEED |
| GLR | GEOTHERMAL LOOP RETURN |
| GLS | GEOTHERMAL LOOP SUPPLY |
| HWR | HEATING WATER RETURN |
| HWS | HEATING WATER SUPPLY |
| SMR | SNOWMLET HEATING RETURN |
| SMS | SNOWMLET HEATING SUPPLY |

### STEAM SYSTEMS

| | | |
|---|---|---|
| HPR | CONDENSATE RETURN | - HIGH PRESSURE |
| HPS | STEAM SUPPLY | - HIGH PRESSURE |
| LPR | CONDENSATE RETURN | - LOW PRESSURE |
| LPS | STEAM SUPPLY | - LOW PRESSURE |
| MPR | CONDENSATE RETURN | - MEDIUM PRESSURE |
| MPS | STEAM SUPPLY | - MEDIUM PRESSURE |
| PC | PUMP CONDENSATE | |

## PLAN SYMBOLS LEGEND

| PLAN | DESCRIPTION |
|---|---|
| # | DRAWING FLAG NOTE |
| # | DEMOLITION DRAWING FLAG NOTE |
| # | REVISION CLOUD TAG |
| | SYSTEM DISCONNECT SYMBOL |
| | SYSTEM NEW CONNECTION SYMBOL |
| XXX ### | MECHANICAL EQUIPMENT CALLOUT |
| XXX | MISC./OWNER EQUIPMENT CALLOUT |
| XXX ### | MEDICAL EQUIPMENT CALLOUT |
| - | ELEVATION/SECTION CALLOUT |

## DUCTWORK LEGEND

| ROUND DUCT 3D | ROUND DUCT PLAN | RECTANGULAR DUCT 3D | RECTANGULAR DUCT PLAN | DESCRIPTION |
|---|---|---|---|---|
| | | | | DUCT RISER |
| | | | | DUCT DROP |
| | | | | 90° ELBOW DOWN (NEGATIVE PRESSURE) |
| | | | | 90° ELBOW DOWN (POSITIVE PRESSURE) |
| | | | | 90° ELBOW UP (NEGATIVE PRESSURE) |
| | | | | 90° ELBOW UP (POSITIVE PRESSURE) |
| | | | | CHANGE IN ELEVATION |
| | | | | SIZE TRANSITION |
| | | | | RIGID DUCT TO FLEX DUCT CONNECTION |
| | | | | 90° RADIUS ELBOW |
| N/A | N/A | | | 90° MITERED ELBOW W/ TURNING VANES |
| | | | | 90° STRAIGHT TEE |
| | | | | 90° CONICAL TEE |
| | | | | 45° BRANCH |
| | | | | 45° CONICAL BRANCH |
| | FD | | FD | FIRE DAMPER |
| | SD | | SD | SMOKE DAMPER |
| | FSD | | FSD | FIRE/SMOKE DAMPER |
| | | | | MANUAL BALANCING DAMPER |
| | MD | | MD | MOTORIZED BALANCING DAMPER |
| | BD | | BD | BACKDRAFT DAMPER |

## PIPING LEGEND

| PLAN | DESCRIPTION |
|---|---|
| **PIPE / PIPE FITTINGS / PIPE CONNECTIONS** | |
| | PIPE UP |
| | PIPE DN |
| | PIPE TEE |
| | PIPE ANCHOR |
| | PIPE CAP |
| | PIPE EXPANSION JOINT |
| | PIPE FLEXIBLE CONNECTION |
| | PUMP / EQUIPMENT CONNECTOR |
| **VALVES / ACCESSORIES** | |
| | AIR VENT |
| | BUTTERFLY VALVE |
| | CHECK VALVE |
| | DOUBLE CHECK BACKFLOW PREVENTER |
| | DRAIN VALVE WITH HOSE END |
| FS | FLOW SWITCH |
| | GATE VALVE |
| | GLOBE VALVE |
| | ISOLATION BALL VALVE |
| | PLUG VALVE |
| | FLOW BALANCING VALVE |
| | PRESSURE GAUGE |
| | PRESSURE REDUCING STATION |
| | PRESSURE TEMPERATURE TAP |
| | SAFTY RELIEF VALVE |
| S | SENSOR (MISCELLANEOUS, REFER TO PLANS) |
| | SOLENOID VALVE |
| | SHOCK ABSORBER |
| | STRAINER |
| | STRAINER WITH BLOW-OFF VALVE |
| | STRAINER WITH VALVE |
| | 2-WAY TEMPERATURE CONTROL VALVE |
| | 3-WAY TEMPERATURE CONTROL VALVE |
| TC | TEMPERTURE CONTROLLER OR SENSOR |
| | THERMOMETER |
| | UNION |
| | VACUUM BREAKER |
| | VENTURI / FLOW INDICATOR |
| **FIXTURES / DRAINS / CLEANOUTS /TRAPS** | |
| | FLOOR DRAIN |
| | FLOOR SINK |
| | FLOOR CLEANOUT |
| | HORIZONTAL CLEANOUT |
| | HOSE BIBB OR WALL HYDRANT |
| | ROOF DRAIN |
| | STEAM TRAP (FT - FLOAT & THERMOSTATIC; TD - THERMODYNAMIC; IB - INVERTED BUCKET; TS - THERMOSTATIC; BP - BALANCED PRESSURE) |
| | STEAM TRAP TEST CHAMBER |

## ABBREVIATIONS

| | |
|---|---|
| AAV | AIR ADMITTANCE VALVE |
| ADA | AMERICANS WITH DISABILITIES ACT |
| AF/AT | AMP FRAME, AMP TRIP |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| AIC | AMPERES INTERRUPTING CAPACITY (SYMMETRICAL) |
| AMP or A | AMPERE |
| AS/AF | AMP SWITCH, AMP FUSE |
| ATS | AUTOMATIC TRANSFER SWITCH |
| AVG | AVERAGE |
| AWG | AMERICAN WIRE GAUGE |
| BAS | BUILDING AUTOMATION SYSTEM |
| BFG | BELOW FINISHED GRADE |
| C | CONDUIT |
| CB | CIRCUIT BREAKER |
| CIRC, CKT | CIRCUIT |
| CT | CURRENT TRANSFORMER |
| (D) or DEMO | DEMOLISH |
| DISC | DISCONNECT |
| (E) | EXISTING |
| EF | EXHASUT FAN |
| EMT | ELECTRICAL METALLIC TUBING |
| ENT | ELECTRICAL NON-METALLIC TUBING |
| EPO | EMERGENCY POWER OFF |
| (F) | FUTURE |
| FA | FIRE ALARM |
| FCT | FOR CONTINUATION |
| FFI | FOR FURTHER INFORMATION |
| FLA | FULL LOAD AMPS |
| FSD | FIRE SMOKE DAMPER |
| FT or ' | FEET |
| GFCI | GROUND FAULT CIRCUIT INTERRUPTER |
| GFP | GROUND FAULT PROTECTION |
| GRD | GROUND |
| W,D,H,L | WIDTH, DEPTH, HEIGHT, LENGTH |
| HOA | HAND-OFF-CUT |
| HP | HORSEPOWER |
| HVAC | HEATING, VENTILATION, AND AIR CONDITIONING |
| IG | ISOLATED GROUND |
| IN or " | INCHES |
| JBOX | JUNCTION BOX |
| KVA | KILOVOLT AMPERES |
| KW | KILOWATT |
| KWH | KILOWATT HOUR |
| LF | LINEAR FEET |
| LTG or LTS | LIGHTING |
| M | METER |
| MAX | MAXIMUM |
| MCA | MINIMUM CIRCUIT AMPACITY |
| MCB | MAIN CIRCUIT BREAKER |
| MCC | MOTOR CONTROL CENTER |
| MCM | THOUSAND CIRCULAR MILS |
| MCP | MOTOR CIRCUIT PROTECTOR |
| MFR | MANUFACTURER |
| MIN | MINIMUM |
| MISC | MISCELLANEOUS |
| MLO | MAIN LUGS ONLY |
| MOCP | MAXIMUM OVERCURRENT PROTECTION |
| MON | MONITORING |
| MTD | MOUNTED |
| (N) | NEW |
| NC / NO | NORMALLY CLOSED / NORMALLY OPEN |
| NEC | NATIONAL ELECTRIC CODE |
| NEMA | NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION |
| NF | NON-FUSED |
| NIC | NOT IN CONTRACT |
| NL | NIGHT LIGHT |
| No. or # | NUMBER |
| NTS | NOT TO SCALE |
| OFCI | OWNER FURNISHED, CONTRACTOR INSTALLED |
| P | POLE |
| PC | PHOTOCELL |
| PH | PHASE |
| PROVIDE | FURNISH AND INSTALL |
| PT | POTENTIAL TRANSFORMER |
| PVC | POLYVINYL CHLORIDE |
| (R) | RELOCATED |
| RE: | REFER TO |
| REC or RECPT | RECEPTACLE |
| RGS | RIGID GALVANIZED STEEL |
| RMS | ROOT MEAN SQUARE |
| SF | SQUARE FEET |
| SPD | SURGE PROTECTION DEVICE |
| SPST | SINGLE-POLE, SINGLE-THROW |
| SQ | SQUARE |
| TC | TIMECLOCK |
| TEL/DATA | TELEPHONE/DATA |
| TV | TELEVISION |
| TYP | TYPICAL |
| UGPS | UNDERGROUND PULL SECTION |
| UON | UNLESS OTHERWISE NOTED |
| UPS | UNITERRUPTABLE POWER SYSTEM |
| V | VOLTS |
| VA | VOLT AMPERES |
| VAV | VARIABLE AIR VOLUME |
| VD | VOLTAGE DROP |
| VFD | VARIABLE FREQUENCY DRIVE |
| W | WIRE |
| w/ | WITH |
| WP | WEATHERPROOF |
| XFRM | TRANSFORMER |
| %Z | PERCENTAGE IMPEDANCE |

## MECHANICAL GENERAL NOTES:

1. GENERAL NOTES, LEGENDS AND DETAILS ARE APPLICABLE TO M-SERIES DRAWINGS.
2. PERFORM WORK IN ACCORDANCE WITH APPLICABLE REGULATIONS, CODES AND STANDARDS, INCLUDING BUT NOT LIMITED TO, LOCAL, STATE AND FEDERAL REGULATIONS, CODES AND STANDARDS.
3. WORK AND EQUIPMENT SHOWN IS DIAGRAMMATIC AND IN APPROXIMATE LOCATIONS. DETERMINE AND COORDINATE EXACT LOCATIONS OF SYSTEMS, EQUIPMENT AND COMPONENTS IN THE FIELD.
4. PROVIDE NEW MATERIALS AND EQUIPMENT UNLESS SPECIFICALLY INDICATED OTHERWISE.
5. COORDINATE WORK OF THIS DIVISION WITH THAT OF OTHER DIVISIONS.
6. SEE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS PERTAINING TO THESE DOCUMENTS.
7. PROVIDE INFORMATION AND HARDWARE AS NECESSARY TO COORDINATE CONCRETE PADS AND STEEL PLATFORMS REQUIRED FOR WORK OF THIS DIVISION.
8. FOR FLOOR MOUNTED MECHANICAL EQUIPMENT, PROVIDE 4-INCH TALL CONCRETE HOUSEKEEPING PADS THAT EXTEND A MINIMUM OF 4-INCHES BEYOND THE EQUIPMENT FOOTPRINT ON EACH SIDE.
9. COORDINATE EXTERIOR WALL PENETRATIONS WITH WORK OF OTHER DIVISIONS AND WITH PROJECT REQUIREMENTS.
10. EXTERIOR LOUVERS ARE INDICATED FOR COORDINATION AND INFORMATION ONLY. DETAILED DESCRIPTIONS AND DIMENSIONS ARE PROVIDED IN THE SPECIFICATIONS AND ARCHITECTURAL DRAWINGS.
11. RUN DUCTS AND PIPING CONCEALED, UNLESS NOTED OTHERWISE. INSTALL DUCTWORK AS CLOSE AS POSSIBLE TO UNDERSIDE OF BEAMS AND JOISTS, UNLESS NOTED OTHERWISE.
12. PROVIDE OFFSETS IN PIPING AND DUCTS (INCLUDING DIVIDED DUCTS AND TRANSITIONS) AROUND OBSTRUCTIONS AT NO ADDITIONAL COST TO THE OWNER.
13. VERIFY EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS AND INSTALLATION INSTRUCTIONS. VERIFY AND PROVIDE DUCT TRANSITIONS TO FURNISHED EQUIPMENT. FIELD VERIFY AND COORDINATE DIMENSIONS BEFORE FABRICATION.
14. PROVIDE ACCESS PANELS WHERE REQUIRED TO SERVICE DAMPERS, HEATERS, VALVES, MOTORS, AND CONCEALED MECHANICAL EQUIPMENT.
15. SUPPORT EQUIPMENT, PIPING AND DUCTWORK FROM BUILDING STRUCTURE AND PROVIDE A VIBRATION-FREE INSTALLATION. NOTIFY ENGINEER OF WEIGHTS AND METHODS OF SUPPORT.
16. SECURELY INSTALL SHEET METAL DUCTWORK AND FITTINGS, WITH REINFORCEMENT AND SUPPORTED IN ACCORDANCE WITH THE LATEST SMACNA STANDARDS.
17. ALL CONTROL AND BAS WIRING SHALL COMPLY WITH NEC AND DIVISION 26.
18. PROVIDE HANGERS, CLAMPS, OFFSETS, EXPANSION JOINTS, ANCHORS AND GUIDES AS NECESSARY TO PREVENT STRESS ON PIPING.
19. PROVIDE VENTS AT ALL HIGH POINTS IN PIPING SYSTEMS AND DRAIN VALVES AT ALL LOW POINTS.
20. PROVIDE AT LEAST THREE ELBOW SWING FOR PIPING TAKE-OFFS TO RISERS AND EQUIPMENT.
21. ALL EXISTING BUILDING CONSTRUCTION, DUCTWORK, PIPING, CONDUITS, ETC. ARE NOT SHOWN ON THE DRAWINGS. BE RESPONSIBLE FOR ROUTING THE NEW DUCTWORK AND PIPING WITH ALL NECESSARY DROPS, OFFSETS, RISERS AND TRANSITIONS AS REQUIRED FOR A COMPLETE INSTALLATION.
22. EQUIPMENT SHUTDOWNS OR CONNECTIONS TO EXISTING EQUIPMENT SHALL NOT BE MADE WITHOUT PRIOR APPROVAL OF THE OWNER.
23. NEW WORK SHALL NOT ENCROACH UPON SERVICE ACCESS AND CLEARANCES FOR EXISTING EQUIPMENT.
24. CONFIRM THE SIZE OF CONNECTIONS TO EXISTING EQUIPMENT, DUCTWORK AND PIPING IN THE FIELD PRIOR TO FABRICATION.
25. PROVIDE TEMPORARY PARTITIONS FOR DUST CONTROL AND TO SEPARATE WORK AREAS FROM OCCUPIED AREAS.
26. PROVIDE DUST PROTECTION FOR ALL NEW AND EXISTING TO REMAIN DUCTWORK AND EQUIPMENT.
27. EXISTING DUCTWORK AND AIR HANDLING EQUIPMENT SHALL NOT BE USED TO SERVE THE WORK AREA DURING CONSTRUCTION WITHOUT WRITTEN PERMISSION FROM THE OWNER. IF THEIR USE IS ALLOWED, PROVIDE CONSTRUCTION FILTERS AT ALL RETURN AND EXHAUST AIR OPENINGS AND WITHIN AIR HANDLING EQUIPMENT. AT COMPLETION OF WORK, ALL TEMPORARY FILTERS SHALL BE REMOVED AND NEW FILTERS INSTALLED.
28. ALL MEANS OF EGRESS SHALL BE MAINTAINED FREE AND CLEAR OF OBSTRUCTIONS AND EXISTING LIFE SAFETY SYSTEMS SHALL REMAIN OPERATIONAL AT ALL TIMES.
29. DUCTWORK, PIPING, AND EQUIPMENT INDICATED TO BE REMOVED, SHALL BE REMOVED IN THEIR ENTIRETY, INCLUDING ALL ASSOCIATED WIRING AND SUPPORTS.
30. ALL MATERIAL INDICATED TO BE REMOVED, INCLUDING LIQUIDS, SHALL BECOME THE PROPERTY OF THE CONTRACTOR UNLESS OTHERWISE NOTED, AND SHALL BE REMOVED FROM THE SITE. REMOVALS AND DISPOSAL SHALL COMPLY WITH THE RULES AND REGULATIONS OF ALL AUTHORITIES HAVING JURISDICTION.
31. COORDINATE REMOVALS WITH ALL OTHER TRADES AND WITH ANY REMEDIATION ACTIVITIES.
32. REMOVE PIPING AND DUCTWORK BACK TO THE POINTS INDICATED AND CAP. WHERE NO SPECIFIC POINT IS INDICATED, REMOVE BACK TO THE NEAREST MAIN DUCT OR PIPE.
33. WHERE REMOVALS EXTEND THROUGH THE ROOF OR EXTERIOR WALL, PROVIDE TEMPORARY WEATHER PROTECTION OF THE OPENING AFTER REMOVAL.
34. PROTECT ITEMS IDENTIFIED FOR SALVAGE FROM WEATHER AND PHYSICAL DAMAGE. COORDINATE DELIVERY OF THESE ITEMS TO THE OWNER.

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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION

LOOKOUT MOUNTAIN, 900 COLORDOW RD, GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO. | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

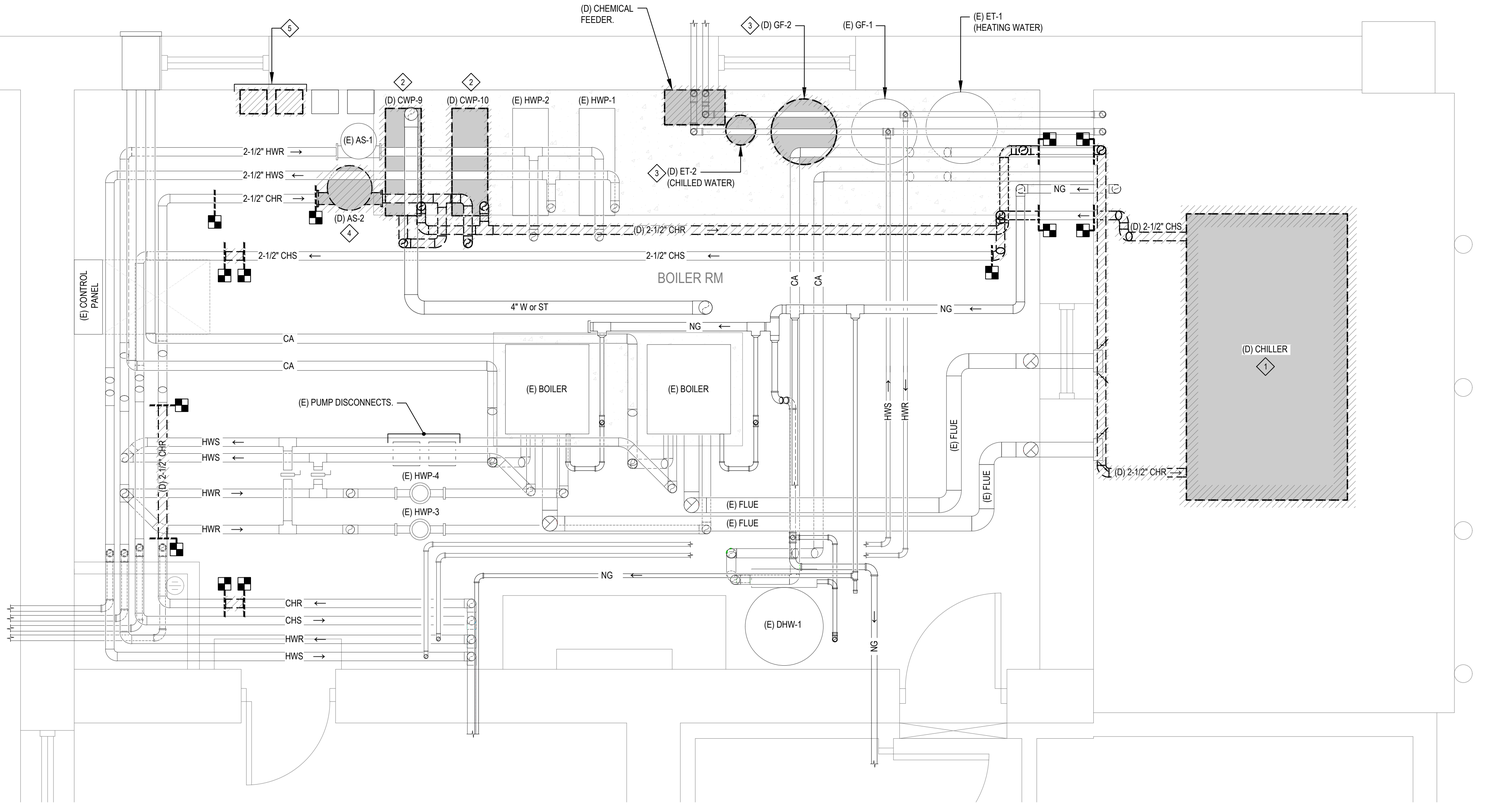
SHEET NAME

MECHANICAL COVER SHEET

SHEET NO.

M-000


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1 M101 PARTIAL BASEMENT MECHANICAL DEMOLITION WORK PLAN

SCALE: 1/2" = 1'-0"

0 5' 10'

## MECHANICAL PRE-TAB NOTES:

1. THE PROJECT CONSISTS OF MODIFICATION, EXTENSION, REPLACEMENT AND/OR RE-USE OF EXISTING HVAC HYDRONIC SYSTEMS. PRIOR TO ANY DEMOLITION PRESCRIBED HEREIN, A PRELIMINARY TAB EXERCISE SHALL BE PERFORMED TO DETERMINE EXISTING CONDITIONS THAT CANNOT OTHERWISE BE VERIFIED THROUGH SIMPLE VISUAL INSPECTION. REFER TO SECTION 23 0593 -TESTING, ADJUSTING, AND BALANCING FOR DETAILED INFORMATION REGARDING PROCEDURES AND PROTOCOLS SPECIFIC TO THE TYPE OF EQUIPMENT REQUIRING PRELIMINARY TAB EXERCISE.

2. THE FIELD PORTION OF THE PRE-TAB EXERCISE SHALL CONSIST OF THE TESTING COMPONENT ONLY, AS SPECIFIED IN SECTION 23 0593 - TESTING, ADJUSTING AND BALANCING. ADJUSTMENT AND BALANCING OF EXISTING SYSTEMS ARE NOT REQUIRED AS PART OF THE PRE-TAB SCOPE. UPON COMPLETION OF THE FIELD PORTION OF PRE-TAB EXERCISE, COMPLETE AND SUBMIT TO ENGINEER FOR REVIEW A CERTIFIED TAB REPORT OF ALL TESTS AND MEASUREMENTS TAKEN.

3. SYSTEMS TO BE INVESTIGATED DURING PRE-TAB EXERCISE:

   A. CHILLER AND CHILLED WATER SYSTEM PUMPS
      A.A. CHILLER NAME PLATE DATA
      A.B. CHILLER MINIMUM FLOW RATING
      A.C. CHILLED WATER LOOP SUPPLY AND RETURN TEMPERATURES
      A.D. DIFFERENTIAL PRESSURE ACROSS PUMPS
      A.E. PUMP GPM
      A.F. SUCTION AND DISCHARGE PRESSURE
      A.G. CALCULATED PUMP HEAD
      A.H. CONTROL VALVE POSITIONS AND FUNCTION AT FCU COILS.

## MECHANICAL DEMOLITION NOTES:

1. EXISTING CONDITIONS DEPICTED IN THESE DRAWINGS ARE DERIVED FROM ORIGINAL DESIGN DRAWINGS, RENOVATION DRAWINGS, AND/OR VARIOUS VERSIONS OF CONTRACTOR AS-BUILT AND DESIGN TEAM RECORD DRAWINGS. WHILE A CONCERTED EFFORT HAS BEEN MADE BY THE ENGINEER TO FIELD VERIFY EXISTING CONDITIONS, THE ENGINEER CANNOT VERIFY THE ABSOLUTE ACCURACY OF THE EXISTING CONDITIONS DEPICTED HEREIN. INDIVIDUALS AND ENTITIES UTILIZING THESE DRAWINGS FOR ANY PURPOSE, INCLUDING BIDDING AND/OR CONSTRUCTION, SHALL BE RESPONSIBLE TO INDEPENDENTLY VERIFY EXISTING CONDITIONS AS THEY RELATE TO THEIR RESPECTIVE SPECIALTY OR DISCIPLINE.

2. THE EXISTING BUILDING SHALL REMAIN IN CONTINUOUS AND NON-INTERRUPTED OPERATION DURING REMODELING/ALTERING OF THE EXISTING BUILDING. THE SPECIFIC AREA(S) BEING REMODELED/ALTERED AT ANY SCHEDULED TIME ARE OBVIOUSLY EXCLUSIVE OF THIS STATEMENT. SERVICES TO EXISTING BUILDING SHALL BE KEPT ON CONTINUOUS OPERATION INCLUDING PLUMBING SERVICES, HVAC SERVICES AND POWER SERVICES STEAM, HEATING, CHILLED, SUPPLY, RETURN, EXHAUST, RELIEF, AND OUTSIDE AIR, ETC. ANY ABSOLUTELY NECESSARY INTERRUPTION OF THESE SERVICES TO ACCOMPLISH PROJECT CONSTRUCTION, SHALL BE ARRANGED WITH THE OWNER THROUGH THE GENERAL CONTRACTOR, A MINIMUM OF TWO (2) WEEKS IN ADVANCE. TEMPORARY SERVICES SHALL BE FURNISHED WHERE NECESSARY TO ACCOMPLISH THIS PURPOSE. TEMPORARY SERVICES SHALL BE REMOVED ONLY AFTER NEW PERMANENT SERVICES ARE INSTALLED AND FULLY OPERATIONAL.

3. THIS CONTRACTOR SHALL BE RESPONSIBLE FOR HIS OWN DEMOLITION, REMOVAL, CAPPING, STORING, ABANDONING, DISCONNECTING, RELOCATING AND RECONNECTION OF EXISTING EQUIPMENT AND MATERIAL. ALL CUTTING, PATCHING, REPAIRING, REPLACEMENT AND REFINISHING, SHALL MATCH THE EXISTING CONSTRUCTION AS NEARLY AS POSSIBLE AND SHALL RESULT IN A COMPLETE, INSULATED, AND OPERATIONAL SYSTEM.

4. EXCEPT WHERE OTHERWISE SHOWN OR NOTED ON DRAWING - "TO BE RETAINED, RELOCATED" OR HEREINAFTER NOTED, ALL EXISTING EQUIPMENT AND MATERIAL IN AREAS TO BE REMODELED/ALTERED SHALL BE REMOVED WHERE THEY INTERFERE WITH PROPOSED NEW CONSTRUCTION AND/OR INTERFERE WITH PROPOSED USAGE OF SPACE BY OWNER AS FOLLOWS:

   A. REMOVE ANY DUCTS PROTRUDING ABOVE FINISHED FLOOR OR THROUGH WALL, CAP AND SEAL, AND FINISH OVER WITH MATERIAL TO MATCH EXISTING.

   B. REMOVE ALL AIR DEVICES AND DAMPERS AS NOTED. CAP AT NEAREST ACTIVE MAIN. REMOVE ABANDONED DUCTWORK BACK TO MAINS.

   C. IN REMODELED/ALTERED AREAS ANY DUCTWORK PASSING THROUGH THE REMODELED AREAS TO SERVE (OR BEING SERVED FROM EXISTING ADJACENT, REMOTE, OR SURROUNDING AREA THAT ARE TO REMAIN) SHALL BE RETAINED AND KEPT OPERATIONAL AND SHALL BE REROUTED IN ALL CASES WHERE THEY INTERFERE WITH ANY NEW WORK OR USAGE TO BE ACCOMPLISHED IN THE REMODELED AREA.

   D. PENETRATIONS THROUGH EXISTING WALLS AND FLOORS FORMERLY OCCUPIED BY REMOVED DUCTWORK SHALL BE PATCHED TO MATCH EXISTING CONSTRUCTION.

5. THIS CONTRACTOR SHALL REFER TO ARCHITECTURAL DRAWINGS TO FAMILIARIZE THEMSELVES WITH EXTENT OF ALTERATION/REMODELING WORK AND MORE SPECIFICALLY NOTE WHERE NEW PARTITIONING IS BEING INSTALLED, WHERE EXISTING PARTITIONING IS BEING REMOVED, WHERE CEILINGS ARE BEING REMOVED AND OR REPLACED, ETC.

6. CONTROL WIRING SHALL BE DEMOLISHED EQUIPMENT (INCLUDING ANY DEMOLISHED ACCESSORIES SUCH AS VALVES) SHALL BE DEMOLISHED BACK TO CONTROL PANEL. SPLICING CONTROL WIRING TO EXTEND TO NEW EQUIPMENT / DEVICES IS NOT ACCEPTABLE. EXISTING CONDUIT MAY BE RE-PURPOSED FOR NEW EQUIPMENT / DEVICES.

## MECHANICAL DEMOLITION KEYNOTES:

1. DEMOLISH EXISTING CHILLER AND ASSOCIATED APPURTENANCES. DEMOLISH EXISTING SUPPLY AND RETURN PIPING TO EXTENT SHOWN AND CAP FOR FUTURE CONNECTION. DEMOLISH EXISTING HEAT TRACE ON SUPPLY AND RETURN PIPING, REFER TO ELECTRICAL DRAWINGS FOR MORE INFORMATION.
2. DEMOLISH EXISTING CHILLED WATER PUMPS AND ASSOCIATED APPURTENANCES. DEMOLISH CHILLED WATER PIPING BACK TO EXTENT SHOWN AND CAP FOR FUTURE CONNECTION.
3. DEMOLISH EXISTING GLYCOL FEEDER AND EXPANSION TANK. DISCONNECT AND CAP EXISTING PIPING FOR FUTURE CONNECTIONS.
4. DEMOLISH EXISTING AIR/DIRT SEPARATOR.
5. DEMOLISH EXISTING PUMP STARTERS. REFER TO ELECTRICAL DRAWINGS FOR MORE INFORMATION.


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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLORADO RD,
GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
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| 1 | 100% CD SET | 09/22/26 |

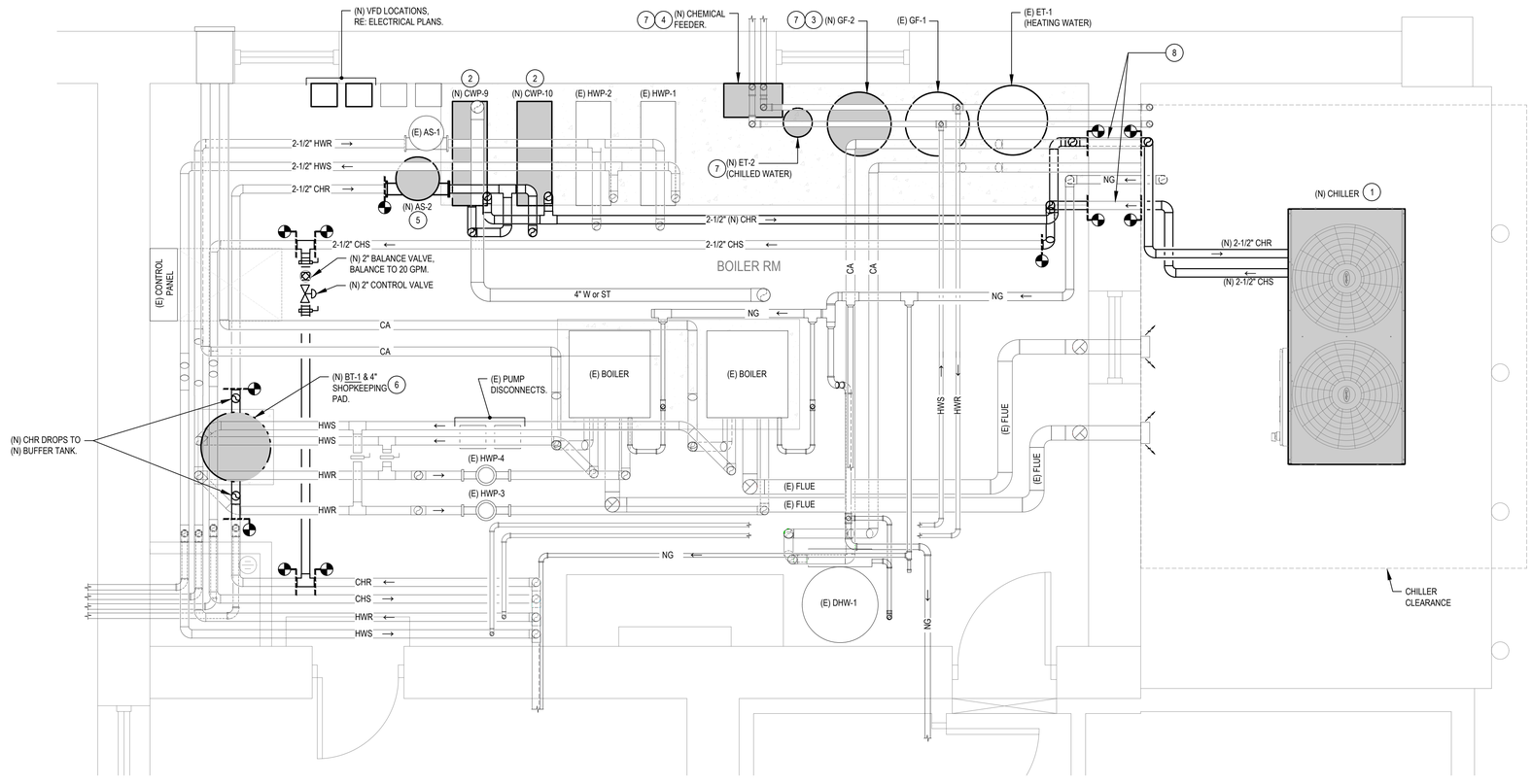
SHEET NAME

PARTIAL BASEMENT MECHANICAL DEMOLITION PLAN

SHEET NO.

M-101


24x36 Z:\Box\STB Projects\2864.18 - Jeffco - Boettcher Mansion Equipment Replacement\600_Drawings\Chiller\M101_Demo.dwg
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1 M201 PARTIAL BASEMENT MECHANICAL NEW WORK PLAN

SCALE: 1/2" = 1'-0"

0 5' 10'

## NOTES:

1. RE: M-000 FOR MECHANICAL SYSTEM NOTES.
2. RE: M-700 FOR MECHANICAL SCHEDULES.
3. PROVIDE NEW CONTROL WIRING FOR ALL NEW EQUIPMENT / DEVICES (INCLUDING ACCESSORIES SUCH AS VALVES). WIRING SHALL BE INSTALLED IN CONDUIT. SPLICING EXISTING CONTROL WIRING TO EXTEND TO NEW EQUIPMENT / DEVICES IS NOT ACCEPTABLE. EXISTING CONDUIT MAY BE RE-PURPOSED FOR NEW EQUIPMENT / DEVICES. PROVIDE NEW CONTROL PANEL / EXPANSION MODULES AS NECESSARY TO COORDINATE WITH NEW EQUIPMENT REQUIRED CONTROL POINTS AND EXISTING BAS.
4. INSTALLATION OF ALL PIPE ACCESSORIES (VALVES, SENSORS, ETC.) SHALL COMPLY WITH MANUFACTURER'S REQUIREMENTS REGARDING MINIMUM STRAIGHT PIPING LENGTHS, DISTANCES FROM ELBOWS. ETC.

## MECHANICAL KEYNOTES:

1. PROVIDE NEW CHILLER ACC-1. REFER TO EQUIPMENT SCHEDULE FOR CAPACITY AND PERFORMANCE REQUIREMENTS. RE: MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR ADDITIONAL INFORMATION.
2. PROVIDE NEW CHILLED WATER PUMPS HWP-9 AND HWP-10. REFER TO EQUIPMENT SCHEDULE FOR CAPACITY AND PERFORMANCE REQUIREMENTS. REFER TO MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR ADDITIONAL INFORMATION.
3. PROVIDE NEW GLYCOL FEEDER GF-1. BASIS OF DESIGN: AXIOM MODEL SF-100 WITH LOW-LEVEL ALARM PANEL RLA10-1-SAA. CONTRACTOR SHALL PROVIDE MANUFACTURER'S STAND OR CUSTOM TANK STAND. STAND HEIGHT OF MINIMUM OF 2FT AFF. REFER TO MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR ADDITIONAL INFORMATION. REFER TO CONTROLS DRAWING M-600 FOR ADDITIONAL INFORMATION.
4. PROVIDE NEW CHEMICAL POT FEEDER. BASIS OF DESIGN: WESSELS COMPANY MODEL CPFT-5. REFER TO MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR ADDITIONAL INFORMATION.
5. PROVIDE NEW AIR/DIRT SEPARATOR AS-1. REFER TO EQUIPMENT SCHEDULE FOR CAPACITY AND PERFORMANCE REQUIREMENTS. REFER TO MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR ADDITIONAL INFORMATION.
6. PROVIDE NEW BUFFER TANK BF-1. REFER TO SCHEDULE FOR CAPACITY. MAINTAIN REQUIRED CLEARANCES FOR CONTROLS PANEL. LOCATE TANK TO NOT INTERFERE WITH HEATING WATER DROPS AND MAINTAIN ACCESS TO HEATING WATER PIPING FOR FUTURE WORK. PROVIDE HOUSE KEEPING PAD, REFER TO DETAIL FOR MORE INFORMATION.
7. REFER TO PIPING SCHEMATIC FOR PIPING CONNECTIONS AND SIZES FOR FOR THE GLYCOL FEEDER,CHEMICAL POT FEEDER, AND EXPANSION TANK.
8. EXISTING PIPING IN WALL SHALL REMAIN. PROVIDE NEW PIPE CONNECTIONS PRIOR TO PENETRATION BOTH INTERIOR AND EXTERIOR

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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION

LOOKOUT MOUNTAIN, 900 COLORADO RD, GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

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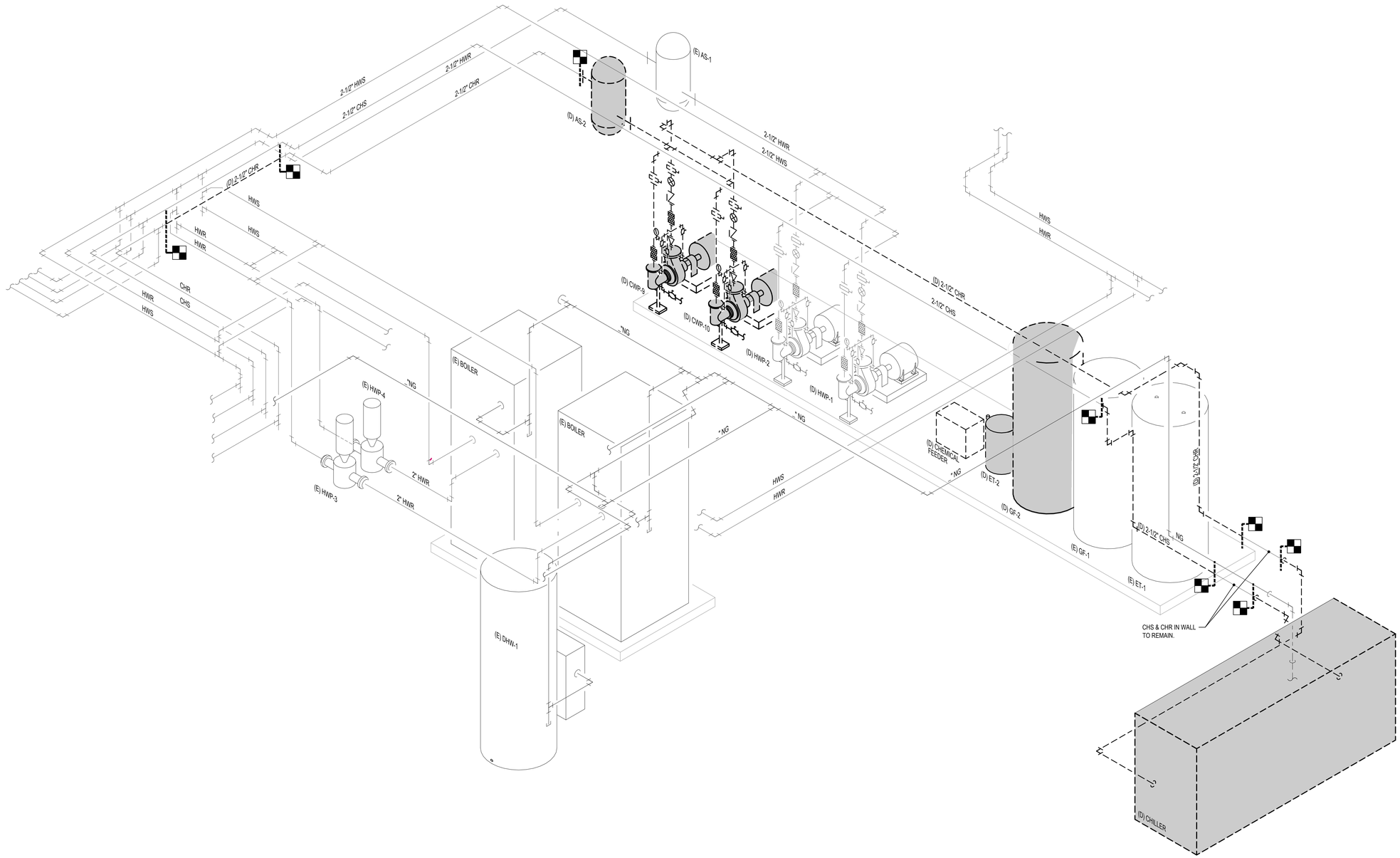
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PARTIAL BASEMENT MECHANICAL NEW WORK PLAN

SHEET NO.

M-201


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1 M301 PIPING ISOMETRIC - DEMOLITION
SCALE: NONE

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# CHILLER REPLACEMENT

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LOOKOUT MOUNTAIN, 900 COLORADO RD,
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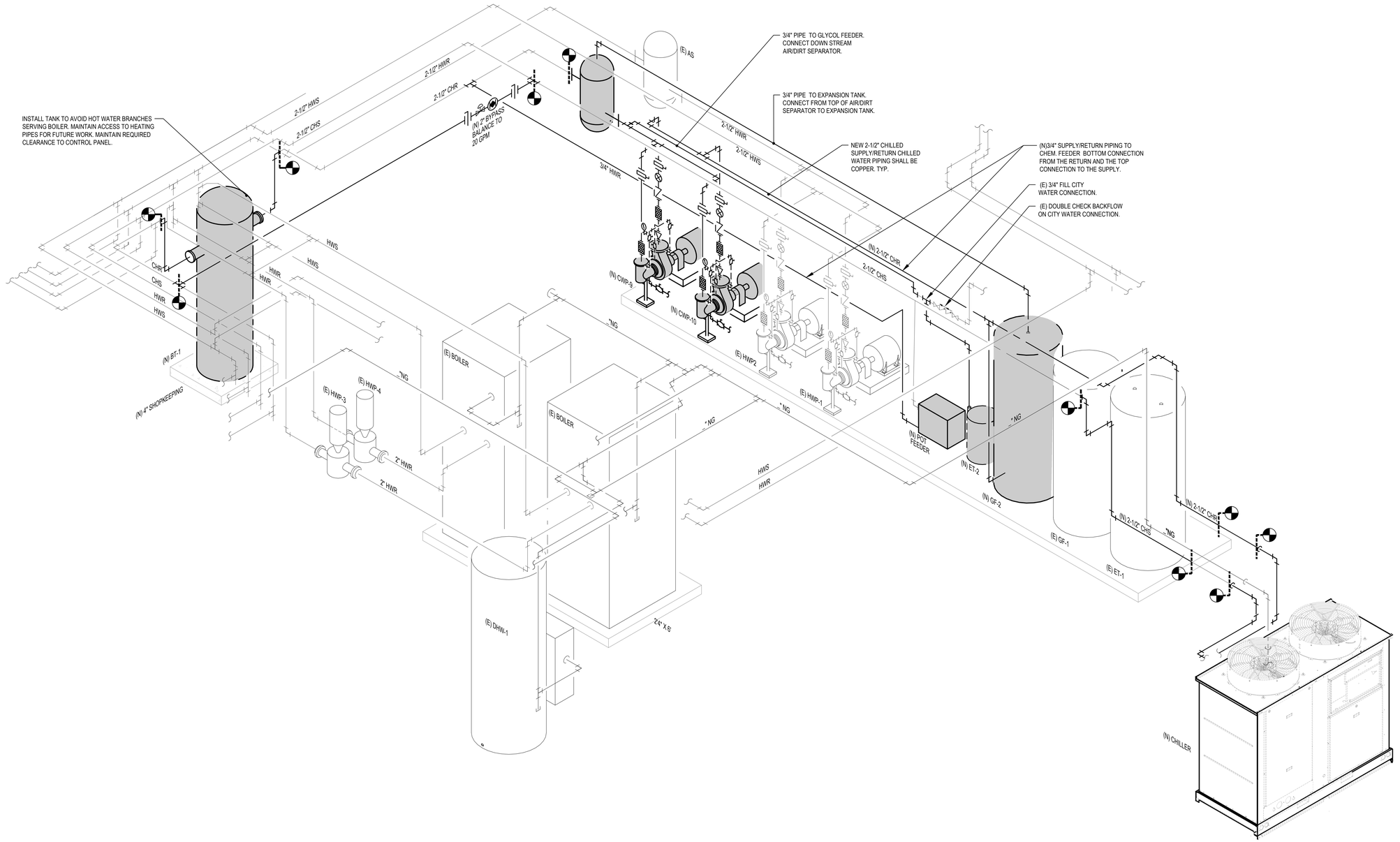
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PIPING ISOMETRIC - DEMOLITION

SHEET NO.
M-301

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GENERAL NOTES:

1. RE: M-500 FOR MECHANICAL DETAILS.
2. RE: M-600 FOR MECHANICAL CONTROLS.
3. RE: M-700 FOR MECHANICAL EQUIPMENT SCHEDULES.




1 M302 PIPING ISOMETRIC - NEW WORK
SCALE: NONE

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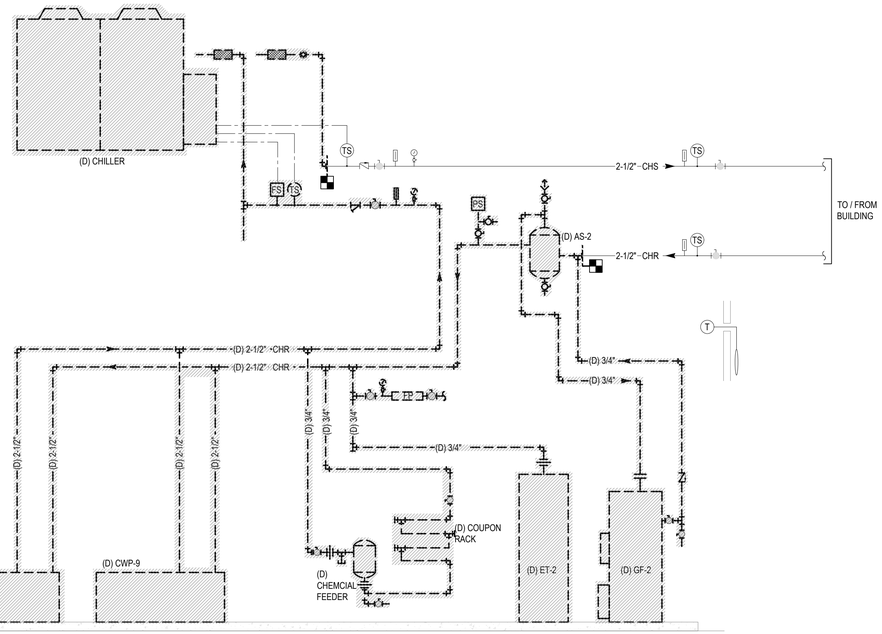
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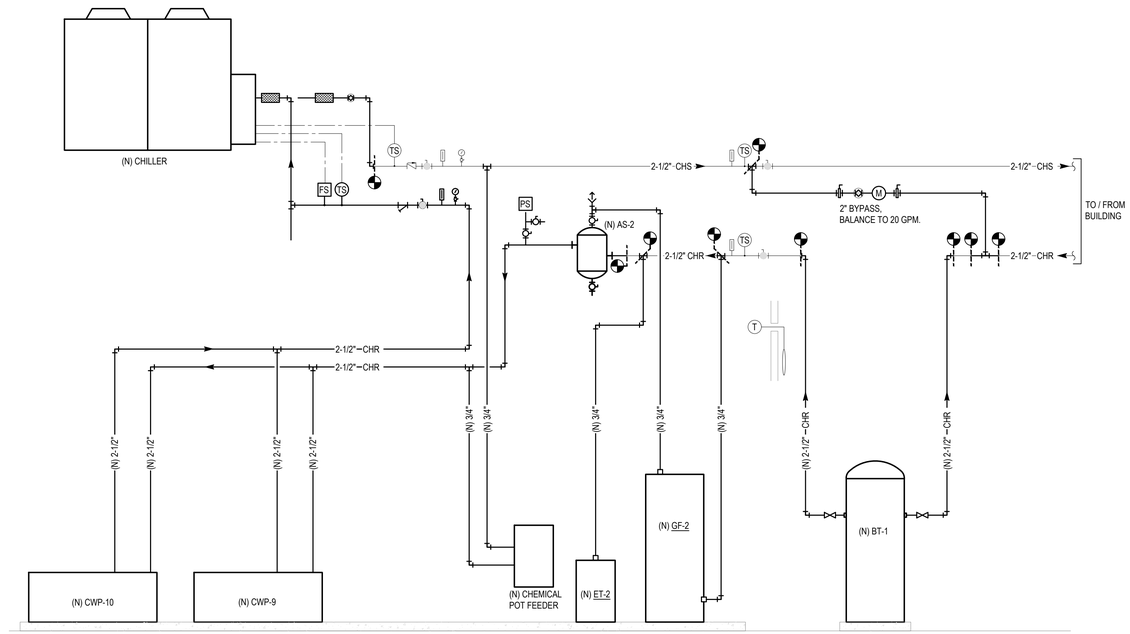
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1 M-400 CHILLED WATER PIPING SCHEMATIC- DEMOLITION
SCALE: NONE




2 M-400 CHILLED WATER PIPING SCHEMATIC - NEW WORK
SCALE: NONE

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# CHILLER REPLACEMENT

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ST+B PROJECT NO.: 2864.18

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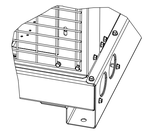
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MECHANICAL SCHEMATIC

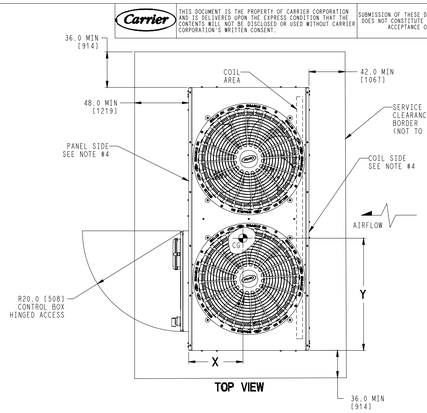
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M-400


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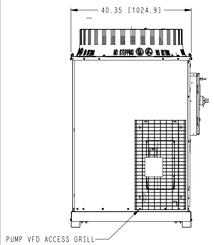
| OPTION | UNIT | NO PUMP CENTER OF GRAVITY | | | SINGLE (1) PUMP NO VFD CENTER OF GRAVITY | | | SINGLE (1) PUMP VFD CENTER OF GRAVITY | | | DUAL (2) PUMP NO VFD CENTER OF GRAVITY | | | UNIT HEIGHT | POWER ENTRY | WATER CONNECTION VICTAULIC |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | X | Y | Z | X | Y | Z | X | Y | Z | X | Y | Z | H | P | WATER IN/OUT |
| GREENSPEED | 30RC020 | 19.37 [492] | 37.32 [948] | 24.89 [632] | 20.12 [511] | 47.01 [1194] | 23.86 [606] | | | | 20.55 [522] | 45.31 [1151] | 23.78 [604] | 66.5 (1689) | 24.9 (631) | 2" |
| | 30RC025 | 19.02 [483] | 35.20 [894] | 25.67 [652] | 20.55 [522] | 49.41 [1255] | 24.69 [627] | | | | 20.91 [531] | 47.80 [1214] | 24.57 [624] | 78.5 (1994) | 36.9 (936) | 2" |
| | 30RC030 | 19.02 [483] | 35.20 [894] | 25.67 [652] | 20.55 [522] | 49.41 [1255] | 24.69 [627] | | | | 20.91 [531] | 47.80 [1214] | 24.57 [624] | 78.5 (1994) | 36.9 (936) | 2" |
| LOW AMBIENT | 30RC020 | 18.86 [479] | 39.09 [993] | 24.06 [611] | 20.51 [521] | 45.20 [1148] | 23.03 [585] | | | | 20.94 [532] | 43.50 [1105] | 23.03 [585] | 66.5 (1689) | 24.9 (631) | 2" |
| | 30RC025 | 18.50 [470] | 36.50 [927] | 23.58 [599] | 20.94 [532] | 47.99 [1219] | 22.76 [578] | | | | 21.30 [541] | 46.34 [1177] | 22.76 [578] | 78.5 (1994) | 36.9 (936) | 2" |
| | 30RC030 | 18.50 [470] | 36.50 [927] | 23.58 [599] | 20.94 [532] | 47.99 [1219] | 22.76 [578] | | | | 21.30 [541] | 46.34 [1177] | 22.76 [578] | 78.5 (1994) | 36.9 (936) | 2" |

NOTES:
1. DO NOT CAP OR OTHERWISE OBSTRUCT THE LIQUID LINE TEMPERATURE RELIEF.
2. Ø7/8 (22.2) PILOT HOLE PROVIDED FOR LOCATING FIELD POWER WIRING. ACTUAL HOLE REQUIRED DEPENDS ON FIELD WIRE SIZING.
3. Ø0.524 (13.30) HOLE USED FOR MOUNTING UNIT.
4. UNIT MUST HAVE CLEARANCES AS FOLLOWS:
   TOP - DO NOT RESTRICT
   COIL SIDE - 42 [1067] FROM SOLID SURFACE.
   PANEL SIDE - 48 [1219] PER NEC.
5. CARRIER DOES NOT RECOMMEND INSTALLATION IN A PIT.
6. UNIT CAN BE HANDLED USING THE FORK TRUCK LIFT POCKETS.
7. WATER CONNECTIONS RECESSED 2-3/8 INCHES INSIDE UNIT. ALL WATER DRAIN AND VENTING HOLES ARE 1/4" NPT.
8. DRAWING SHOWN WITH GREENSPEED OPTION.
9. DIMENSIONS ARE IN "INCH (MM)".

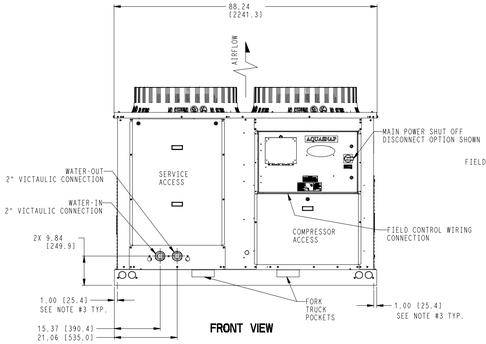


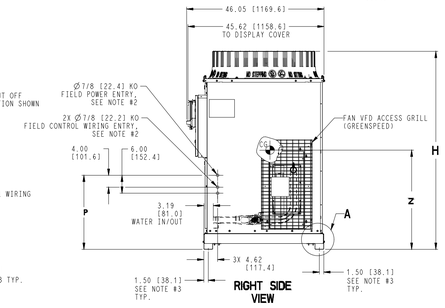

TOP VIEW




LEFT SIDE VIEW



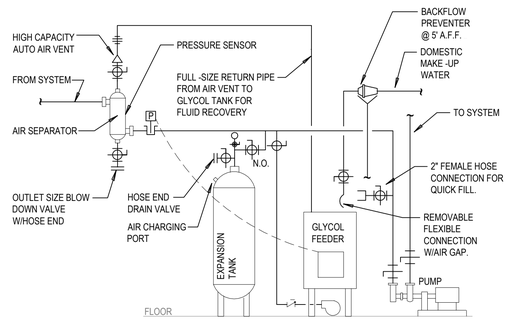

FRONT VIEW



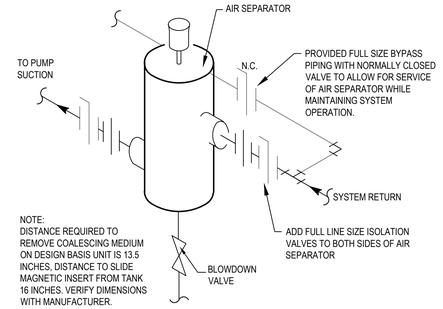

RIGHT SIDE VIEW

| ITC CLASSIFICATION | SHEET | DATE | SUPERCEDES | | |
|---|---|---|---|---|---|
| U.S. ECCN:EAR99 | 1 OF 1 | 09/11/2024 | - | 30RC 020-030 UNIT ASSY | 30RC60001020 REV B |

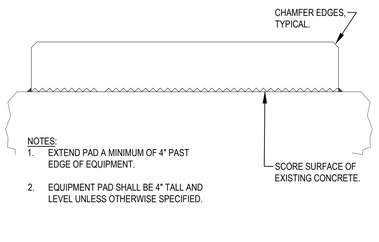
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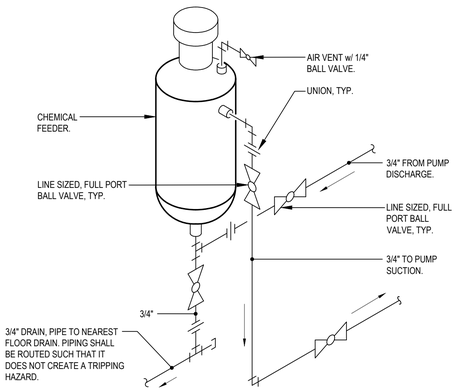

6 M-500 GLYCOL FEEDER / EXPANSION TANK DETAIL
SCALE: NONE



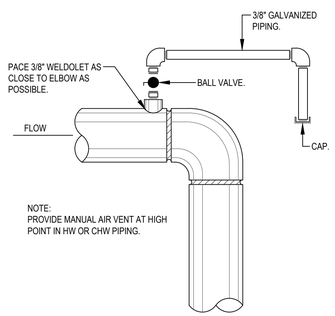

3 M-500 AIR SEPARATOR DETAIL
SCALE: NONE



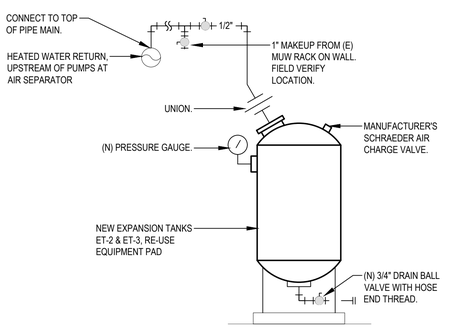

5 M-500 EQUIPMENT PAD DETAIL
SCALE: NONE




2 M-500 CHEMICAL FILL-POT DETAIL - HOT & CHILLED WATER SYSTEMS
SCALE: NONE




4 M-500 MANUAL AIR VENT DETAIL
SCALE: NONE




1 M-500 EXPANSION TANK DETAIL
SCALE: NONE

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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLOROW RD, GOLDEN, CO 80401

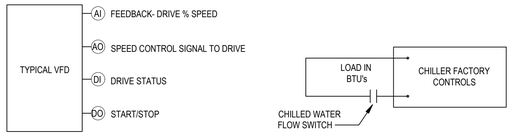
ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
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| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME
MECHANICAL DETAILS

SHEET NO.
M-500

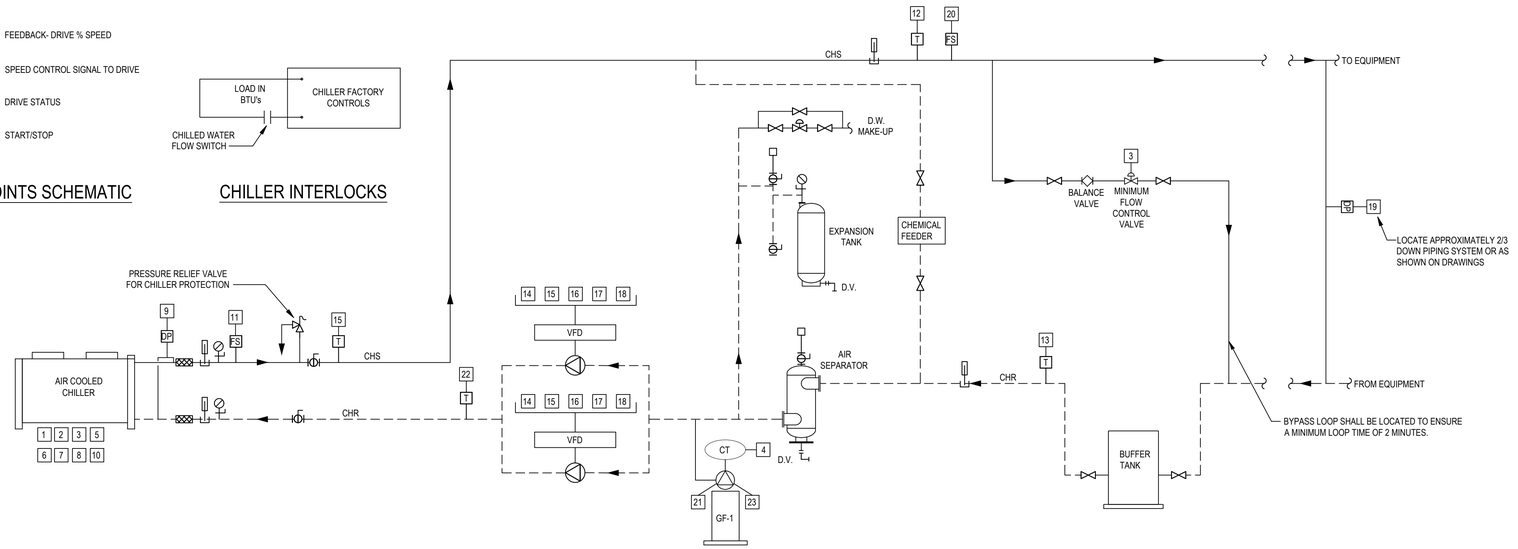
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9/22/2026 9:57 AM




VFD CONTROL POINTS SCHEMATIC




CHILLER INTERLOCKS




SEQUENCE OF OPERATION

GENERAL:

THE EXISTING BAS IS THROUGH RELIABLE CONTROLS

THE CHILLED WATER SYSTEM USES VARIABLE-FLOW PRIMARY DISTRIBUTION WITH AN AIR COOLED CHILLER . CIRCULATION IS PROVIDED WITH TWO VARIABLE SPEED DRIVEN END SUCTION CENTRIFUGAL PUMPS CONFIGURED AS PRIMARY / STAND-BY. THESE PUMPS SHALL OPERATE IN A LEAD / LAG MODE. THE PLANT PROVIDES CHILLED WATER FOR THE RESPECTIVE FAN COIL UNITS (FCU) . THE LOOP SHALL BE MAINTAINED WITH A CHILLER EWT (RETURN WATER) TEMPERATURE OF 55°F (ADJ) AND A CHILLER LWT (SUPPLY WATER) TEMPERATURE OF 45°F (ADJ).

THE CHILLED WATER FLOW RATE THROUGH THE CHILLER EVAPORATOR WILL VARY AS THE CHILLED WATER COOLING COIL CONTROL VALVES VARY THE FLOW RATE TO MEET THE COOLING LOAD.

THE CHILLER BYPASS VALVE AND THE CHILLER DIFFERENTIAL PRESSURE SENSOR MUST BE DIRECTLY CONNECTED TO THE SAME BAS CONTROLLER TO MAINTAIN RAPID RESPONSE TO VARIATIONS IN SYSTEM CHILLED WATER FLOW.

SET THE CHILLED WATER PUMP VFD RAMP UP SPEED AS RECOMMENDED BY THE CHILLER MANUFACTURER TO MAINTAIN THE RATE OF CHANGE OF CHILLED WATER FLOW IN THE SYSTEM WITHIN THE CHILLER MANUFACTURER'S RECOMMENDATIONS.

THE CHILLED WATER SYSTEM IS CONFIGURED TO OPERATE YEAR-ROUND, 24 HOURS PER DAY. THE BAS OPERATOR SHALL BE ABLE TO ENABLE OR DISABLE THE OPERATION OF THE ENTIRE CHILLED WATER PLANT USING A SINGLE TOGGLE OR COMMAND.

THE CHILLER SHALL BE CONTROLLED BY THE CHILLER CONTROL PANEL. THE CONTROL POINTS SHALL BE SENT TO THE BUILDING BAS FOR INFORMATION AND ALARM PURPOSES. THE REST OF THE COMPONENTS ASSOCIATED WITH THE CHILLER PLANT SHALL BE CONTROL BY THE BAS.

THE BAS SHALL SEND A SIGNAL TO CHILLER CONTROL PANEL TO ENABLE CHILLER OPERATION. THE CHILLER CONTROL PANEL SHALL FIRST START THE LEAD CHILLED WATER PUMP AND SLOWLY RAMP IT UP TO SPEED OVER A 30 SECOND PERIOD. THE CHILLED WATER PUMP SPEED SHALL BE VARIED BY THE VARIABLE FREQUENCY DRIVE (VFD) TO MAINTAIN THE REQUIRED DIFFERENTIAL PRESSURE IN THE PIPING SYSTEM THROUGH A CONTROL SYSTEM INTERFACE WITH THE DRIVE. AFTER THE PUMP SPEED HAS STABILIZED, AFTER THE CHILLED WATER FLOW HAS BEEN PROVEN BY THE CHILLER FLOW SWITCH AND ALL INTERNAL SAFETIES HAVE BEEN SATISFIED, THE CHILLER SHALL BE ENABLED AND CHILLED WATER TEMPERATURE SHALL BE MAINTAINED BY THE CHILLER'S INTERNAL ELECTRONIC CONTROL SYSTEM.

CHILLED WATER TEMPERATURE SENSORS SHALL MONITOR SUPPLY AND RETURN WATER TEMPERATURES AT THE CHILLER AND PROVIDE THIS DATA TO THE BAS FOR DISPLAY ON THE GRAPHIC. CHILLED WATER SUPPLY TEMPERATURE SHALL BE MAINTAINED AT 45°F (ADJ.) BY THE CHILLER INTERNAL CONTROL SYSTEM.

IF THE LEAD PUMP SHOULD FAIL TO OPERATE, THE LEAD PUMP SHALL BE DE-ENERGIZED, THE LAG PUMP SHALL BE ENABLED AND AN ALARM SHALL BE SENT TO THE BAS. THE CHILLED WATER PUMP(S) VFD SHALL GRADUALLY INCREASE THE PUMP(S) OUTPUT TO MAINTAIN A MINIMUM PRESSURE DIFFERENTIAL(ADJ) ACROSS THE CHILLER EVAPORATOR COIL. WHEN ALL DIFFERENTIAL PRESSURE SENSORS ACROSS THE COILS INDICATE DIFFERENTIAL PRESSURES ABOVE THE SETPOINTS, THE PUMP(S) VFD SHOULD ADJUST TO GRADUALLY REDUCE PUMP(S) OUTPUT.

THE CHILLED WATER SYSTEM CONTAINS A FIVE (5) 3-WAY CONTROL VALVES. HOWEVER, IN ORDER TO MAINTAIN CHILLER MINIMUM FLOW REQUIREMENTS, A SYSTEM BYPASS SHALL ALSO BE PROVIDED. ONLY WHEN THE LEAD PUMP IS AT THE MINIMUM FLOW FOR A MINIMUM OF 120 SECONDS (ADJ), AND EVAPORATOR DIFFERENTIAL PRESSURE ABOVE SETPOINT (ADJ) FOR ADDITIONAL 60 SECONDS (ADJ), SHALL THE BY-PASS VALVE OPEN MAINTAIN THE SYSTEM DIFFERENTIAL PRESSURE SETPOINT (ADJ). THE DIFFERENTIAL PRESSURE AS SENSED BY SENSOR SHALL BE ADJUSTED DURING TESTING AND BALANCING HIGH ENOUGH TO KEEP THE FARTHEST COIL DIFFERENTIAL PRESSURE SATISFIED. AS FLOW DEMAND INCREASES, THE BY-PASS VALVE SHALL CLOSE PRIOR TO THE LEAD PUMP VFD INCREASING PUMP OUTPUT. THE LEAD PUMP VFD SHALL NOT INCREASE ABOVE MINIMUM VFD SETTING UNLESS THE BYPASS VALVE HAS BEEN IN CLOSED POSITION FOR 90 SECONDS (ADJ). THE MINIMUM PUMP SPEED SHALL BE DETERMINE DURING TAB EFFORTS AND COORDINATED WITH THE PURCHASED CHILLER'S MINIMUM FLOW REQUIREMENT IN CONJUNCTION WITH THE EXISTING SYSTEM 3-WAY VALVES.

THE INITIAL PIPING SYSTEM DIFFERENTIAL PRESSURE CONTROL POINT SHALL BE AS DETERMINED BY THE TESTING AND BALANCING CONTRACTOR. THIS DIFFERENTIAL PRESSURE CONTROL POINT SHALL SERVE AS THE INITIAL AND HIGHEST SETTING. THE BAS GRAPHIC DISPLAY SHALL DISPLAY THE CURRENT PIPING SYSTEM DIFFERENTIAL PRESSURE SETPOINT (WHICH WILL VARY DURING THE COURSE OF THE DAY).

A LINEAR BYPASS VALVE SHALL MAINTAIN ADEQUATE FLOW THROUGH THE CHILLER. THE PRECISION DIFFERENTIAL PRESSURE SENSOR [AT THE CHILLER SHALL SENSE FLUCTUATIONS IN WATER FLOW AND CONTROL TO PREVENT THE EVAPORATOR FLOW RATE FROM BEING REDUCED TOO QUICKLY AS FOLLOWS: ON A FALL IN DIFFERENTIAL PRESSURE BELOW SETPOINT, THE BAS SHALL CALL FOR THE BYPASS VALVE TO OPEN AN INCREASING AMOUNT.

THE CHILLER SHALL HAVE A FLOW SWITCH HARD WIRED TO THE CHILLER CONTROL PANEL INDICATING WATER FLOW. THE BYPASS VALVE SHALL BE CLOSED AT ALL TIMES UNLESS THE EVAPORATOR DIFFERENTIAL PRESSURE SENSOR HAS FALLEN BELOW THE MANUFACTURERS MINIMUM RECOMMENDED SETTING INDICATING LOW EVAPORATOR FLOW.

ON A REDUCTION IN CHILLER FLOW RATE AS DETERMINED BY THE DIFFERENTIAL PRESSURE ACROSS THE EVAPORATOR BELOW THE CHILLER MANUFACTURER'S MINIMUM RECOMMENDED FLOW RATE, THE BYPASS VALVE SHALL QUICKLY ACTUATE TO MAINTAIN ADEQUATE FLOW TO KEEP THE CHILLER ON-LINE. LIKEWISE, SHOULD THE FLOW RATE THROUGH THE CHILLER BE REDUCED AT A RATE OF CHANGE GREATER THAN THE MAXIMUM RECOMMENDED RATE OF THE CHILLER MANUFACTURER, THE BYPASS VALVE SHALL QUICKLY ACTUATE TO MAINTAIN ADEQUATE FLOW TO KEEP THE CHILLER ON-LINE AND AN ERROR MESSAGE SHALL BE SENT TO THE OPERATORS WORKSTATION AS FOLLOWS: "THE RATE OF CHANGE OF FLOW IN THE CHILLED WATER HEAT TRANSFER COILS IS TOO RAPID AND MAY CAUSE THE CHILLER TO DROP OFF-LINE IF NOT CORRECTED. CHECK THE LOOP TUNING.

ON SHUTDOWN OF THE CHILLED WATER PUMP, THE DIGITAL CONTROLLER SHALL CAUSE THE VFD TO SLOW THE PUMP DOWN TO 10% SPEED OVER 60 SECONDS BEFORE TAKING IT OFF-LINE.

THE BAS SHALL RECORD THE NUMBER OF HOURS OPERATION OF ALL PUMPS AND THE CHILLER AND RECORD THOSE HOURS FOR MAINTENANCE RECORDS. THE PUMPS SHALL AUTOMATICALLY SWITCH FROM LEAD TO LAG AND LAG TO LEAD AFTER EVERY 240 HOURS (ADJ) OF PUMP OPERATION AT THE NEXT OPPORTUNE TIME.

AN ADJUSTABLE CURRENT SENSING RELAY AND THE CHILLED WATER PUMP VFD RUN STATUS SHALL PROVIDE PROOF THAT THE PRIMARY CHILLED WATER PUMP IS OPERATING WHEN COMMANDED ON. IF THE CURRENT SENSING RELAY OR VFD RUN STATUS INDICATES THAT THE PUMP IS NOT OPERATING WHEN COMMANDED ON BY THE BAS, AN ALARM SIGNAL SHALL BE SENT TO THE BAS OPERATOR WORKSTATION.

THE BAS SHALL MONITOR THE CHILLED WATER SUPPLY AND RETURN WATER TEMPERATURES. THE BAS SHALL CALCULATE AND DISPLAY THE TOTAL COOLING LOAD AT ANY TIME TO THE BUILDING IN BRITISH THERMAL UNITS (BTU'S) USING THE SYSTEM FLOW AND DISTRIBUTION LOOP TEMPERATURE DIFFERENCE.

THE GLYCOL FEED TANK AND PUMP CONTROL SHALL BE BASED ON THE INTEGRAL PRESSURE SWITCH PROVIDED WITH THE GLYCOL MAKE-UP UNIT. WHEN THE PRESSURE IN THE RESPECTIVE CHILLED WATER LOOP DECREASES BELOW THE PRESSURE SETTING/CUT-IN PRESSURE (SEE SCHEDULE FOR MORE INFORMATION), THE GLYCOL FEED PUMPS SHALL ENERGIZE. THE PUMPS SHALL STOP WHEN THE PRESSURE IS 5 PSI (ADJUSTABLE) ABOVE THE PRESSURE SETTING. A BAS ALARM SHALL BE PROVIDED UPON OPERATION OF THE GLYCOL FEED PUMP FOR A CONTINUOUS ADJUSTABLE PERIOD.

## BAS POINT LIST

| TAG | NAME/FUNCTION | AI | AO | DI | DO | REMARKS | QTY |
|---|---|---|---|---|---|---|---|
| 1 | CHILLER ON/OFF CONTROL | | | | ● | TO CHILLER CONTROL PANEL | 1 |
| 2 | CHILLER CIRCUIT 1 STATUS | | | ● | | FROM CHILLER CONTROL PANEL | 1 |
| 3 | BYPASS CONTROL VALVE | | | ● | | FROM CHILLER CONTROL PANEL | 1 |
| 4 | GLYCOL FEEDER PUMP STATUS | | | ● | | NEW POINT | 1 |
| 5 | CHILLER POWER USAGE | ● | | | | FROM CHILLER CONTROL PANEL | 1 |
| 6 | CHILLER RLA | ● | | | | FROM CHILLER CONTROL PANEL | 1 |
| 7 | CHILLER RUNTIME | ● | | | | FROM CHILLER CONTROL PANEL | 1 |
| 8 | CHILLER FAULT- GENERAL | | | ● | | FROM CHILLER CONTROL PANEL | 1 |
| 9 | EVAPORATOR DIFFERENTIAL PRESSURE | ● | | | | NEW POINT | 1 |
| 10 | CHILLER LOAD | ● | | | | FROM CHILLER CONTROL PANEL | 1 |
| 11 | CHILLER FLOW SWITCH | | | ● | | NEW POINT | 1 |
| 12 | SYSTEM CHS TEMPERATURE | ● | | | | HIGH AND LOW TEMP ALARM | 1 |
| 13 | SYSTEM CHR TEMPERATURE | ● | | | | HIGH AND LOW TEMP ALARM | 1 |
| 14 | CWP PUMP ON/OFF | | | | ● | NEW POINT TO VFD PANEL | 2 |
| 15 | CWP STATUS | | | ● | | NEW POINT | 2 |
| 16 | CWP VFD FAIL | | | ● | | NEW POINT FROM VFD PANEL | 2 |
| 17 | CWP VFD SPEED OUTPUT | | ● | | | NEW POINT TO VFD PANEL | 2 |
| 18 | CWP VFD % FEEDBACK | ● | | | | NEW POINT, FROM VFD PANEL | 2 |
| 19 | SYSTEM DIFFERENTIAL PRESSURE SENSOR | ● | | | | NEW POINT, NEW SENSOR | 1 |
| 20 | CHS FLOW SENSOR | ● | | | | NEW POINT | 1 |
| 21 | GLYCOL FEEDER PUMP ON / OFF | | | | ● | NEW POINT | 1 |
| 22 | CHILLER ENTERING WATER TEMPERATURE | ● | | | | NEW POINT | 1 |
| 23 | LOW LEVEL ALARM | | | ● | | NEW POINT | 1 |

NOTE:

1. PROVIDE TREND DATA FOR ALL POINTS SHOWN. TRENDING SHALL BE PROVIDED AT A MINIMUM OF EVERY 15 MINUTES AND SHALL BE BACKED UP TO A SERVER AUTOMATICALLY EVERY DAY AND KEPT FOR A PERIOD OF 12 MONTHS. EACH TREND FILE SHALL BE APPENDED SUCH A THAT A SINGLE FILE IS PROVIDE FOR EACH MONTH. TREND DATA EXPORTS SHALL BE PROVIDED IN CSV OR XLS FILE FORMAT. THE CONTROL SYSTEM SHALL ALSO BE CAPABLE OF PROVIDING CUSTOM EXPORTS BASED ON SELECTED POINTS AND TIME FRAMES TO THE SAME FILE FORMATS.
2. CHECK THE LOOP TUNING ON THE EXISTING FCU'S THREE-WAY CONTROL VALVES. ENSURE EXISTING THREE -WAY VALVES ARE FULLY OPERATIONAL AND PROVIDES BYPASS FOR RESPECTIVE FCU'S UPON NO DEMAND/CALL FOR COOLING FOR RESPECTIVE ZONE.
3. ALL CONTROL DEVICES SHOWN IN DIAGRAM SHALL BE NEW CONTROL DEVICES. WHILE SOME CONTROL POINTS MAY BE INCLUDED IN THE IN EXISTING CONTROL SYSTEM WITH THE EXISTING/DEMOLISHED EQUIPMENT, CONTROL WIRING AND REPROGRAMMING SHALL STILL BE PROVIDED TO INTEGRATE NEW DEVICES AND PROVIDE OPERATION AS NOTED IN SEQUENCE OF OPERATION.

1 M-600 CHILLED WATER CONTROLS
NO SCALE

STAMP

COLORADO LICENSED 100D SPACER PROFESSIONAL ENGINEER 45310

9/22/2026


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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLORDW RD,
GOLDEN, CO 80401

ST+B PROJECT No.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME
MECHANICAL CONTROLS

SHEET No.
M-600


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## AIR COOLED CHILLER SCHEDULE

| TAG | SERVICE | TYPE | NOMINAL TONNAGE | COOLING CAPACITY (TONS) | AMBIENT DB TEMPERATURE (°F) | EVAPORATOR | FLUID | MINIMUM FLOW (GPM) | FLOW RATE (GPM) | EWT (°F) | LWT (°F) | WATER PRESSURE DROP (FT. HD.) | REFRIGERANT | MINIMUM FULL LOAD EFFICIENCY (BTU/Wh) | MINIMUM PART LOAD EFFICIENCY (IPLV.IP) (BTU/Wh) | FOULING FACTOR ( HR-SF-F) / BTU ) | ELECTRICAL | | | | | | | | BASIS OF DESIGN | | | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | KW | CIRCUITS | MCA | MOCP | VOLTS | PHASE | HZ | EMERGENCY POWER | MANUFACTURER | MODEL | OPERATING WEIGHT, LBM | DIMENSIONS L x W x H (IN) | |
| ACC-1 | CHILLED WATER | SCROLL COMPRESSOR | 30 | 27 | 105 | BRAZED PLATE AND FRAME HX | 30% PROPYLENE GLYCOL | 40 | 67 | 55.00 | 45.00 | 18 | R-32 | 8.12 | 15.63 | 0.0001 | 51.7 | 1 | 143.5 | 200 | 208 | 3 | 60 | NO | CARRIER | 30RC-030 | 1,432 | 88 x 43 x 78 | ALL APPLY |

| REMARKS | |
|---|---|
| 1. | FULL HAIL GUARDS, LOW SOUND KIT (INCLUSIVE OF ACOUSTIC FANS AND COMPRESSOR SOUND BLANKS COMPRESSORS), ENERGY MANAGEMENT MODULE (EMM), ISOLATION VALVES, LOW AMBIENT OPERATION TO -20°F, AND VARIABLE SPEED CONDENSER FANS. |
| 2. | EQUIPMENT SELECTED AT ELEVATION OF 7,500 FT |
| 3. | SINGLE POINT POWER CONNECTION. |
| 4. | TESTED IN ACCORDANCE WITH AHRI STANDARD 550/590. |
| 5. | ALL MOTORS APPLIED WITH VFD SHALL BE PROVIDED WITH PREMIUM EFFICIENCY INVERTER DUTY MOTOR AND SHAFT GROUNDING BRUSH (AEGIS OR EQUIVALENT). |
| 6. | CHILLERS TO BE PROVIDED WITH FACTORY PROVIDED UNIT MOUNTED NON-FUSED DISCONNECT. |
| 7. | PROVIDE DIGITAL COMPRESSOR. |
| 8. | UNIT SHALL HAVE 10 kAIC SCCR . |
| 9. | PROVIDE WITH ALUMINUM FINS AND COPPER TUBE CONDENSER COIL. |
| 10. | PROVIDE WITH EVAPORATOR HEATER AND HEAT TRACING ON EXTERIOR CHILLED WATER PIPING FOR FREEZE PROTECTION. SEE ELECTRICAL DRAWINGS FOR ADDITIONAL INFORMATION ON PIPING HEAT TRACE |
| 11. | COMPRESSORS SHALL BE PROVIDED WITH VIBRATION ISOLATION. |
| 12. | PROVIDE WITH A MINIMUM OF 22 CAPACITY CONTROL STEPS. |

## PUMP SCHEDULE

| TAG | SERVICE | TYPE | FLUID TYPE | FLOW RATE (GPM) | HEAD FEET | RPM | SUCTION DIAMETER INCH | DISCHARGE DIAMETER INCH | CONNECTION SIZE INCH | MIN PUMP EFFICIENCY % | NPSHr FEET | ELECTRICAL | | | | | | | BASIS OF DESIGN | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | BHP | HP | VOLTS | PHASE | HZ | EMERGENCY POWER | VFD (Y/N) | MANUFACTURER | MODEL | |
| CWP-9 | SYSTEM DISTRIBUTION CHILLED WATER | BASE MOUNTED END SUCTION | 30% PROPYLENE GLYCOL | 67 | 85 | 1800 | 2.0 | 1.5 | 2.5 | 49.9 | 5.41 | 2.94 | 5.0 | 208 | 3 | 60 | NO | YES | BELL & GOSSETT | E-1510 | 1, 3, 4, 5, 6, 7, 8 |
| CWP-10 | SYSTEM DISTRIBUTION CHILLED WATER | BASE MOUNTED END SUCTION | 30% PROPYLENE GLYCOL | 67 | 85 | 1800 | 2.0 | 1.5 | 2.5 | 49.9 | 5.41 | 2.94 | 5.0 | 208 | 3 | 60 | NO | YES | BELL & GOSSETT | E-1510 | 1, 3, 4, 5, 6, 7, 8 |

| REMARKS | |
|---|---|
| 1. | PREMIUM EFFICIENCY DUTY MOTOR. |
| 3. | STATED SPEED IS THE MAXIMUM MOTOR SPEED ALLOWED. |
| 4. | PROVIDE SHAFT GROUNDING RINGS FOR EACH BEARING ON MOTORS POWERED THROUGH VARIABLE FREQUENCY DRIVES. |
| 5. | PERFORM PRE-TAB OF EXISTING CHILLED WATER SYSTEM. VERIFY EXISTING FLOW, DEMAND AND PUMP HEAD CRITERIA PRIOR TO PUMP PROCUREMENT. REPORT FINDINGS TO ENGINEER FOR DIRECTION BEFORE PROCUREMENT. |
| 6. | PROVIDE INVERTER DUTY RATED MOTOR. |
| 7. | PROVIDE VFD WITH INTEGRAL DISCONNECT MEETING THE REQUIREMENTS OF DIV. 26 SPECIFICATIONS. |
| 8. | 100% REDUNDANT PUMP. |

## AIR SEPARATOR SCHEDULE

| TAG | TYPE | SERVICES | FLUID | MAX FLOW (GPM) | PRESS. DROP (IN.WC) | PIPE CONNECTION (INCHES) | WEIGHT (LBS) | MANUFACTURER | MODEL NUMBER | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| AS-2 | COALESCING AIR & DIRT SEPARATOR | CHILLED WATER SYSTEM | 30% GLYCOL | 67 | 0.9 | 2.5 | 150 | SPIROTHERM | VDT-250 | 1, 2, 3 |

| REMARKS | |
|---|---|
| 1 | STANDARD VELOCITY RATING WITH REMOVABLE HEAD AND MAGNETIC INSERT. |
| 2 | CONTRACTOR SHALL COORDINATE SERVICE REQUIREMENTS NECESSARY WITH PURCHASED EQUIPMENT AND ENSURE MINIMUM REQUIREMENTS ARE PROVIDED. |
| 3 | ASME RATED. |

## EXPANSION TANK SCHEDULE

| TAG | SYSTEM SERVED | TYPE | TANK PRESSURE RATING (PSI) | SYSTEM TEMPERATURE RANGE (°F) | MIN ACCEPT. VOLUME (GALLONS) | TANK VOLUME (GALLONS) | BASIS OF DESIGN | | DRY / OPT WEIGHT, LBS | ACCESSORIES | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | MFG | MODEL | | | |
| ET-2 | CHILLED WATER SYSTEM | DIAPHRAM | 125 | 40-55 | .66 | 6.4 | AMTROL | AX-15DD | 9 / 70 | AS PER DETAIL 4 / M-500 | 1, 2, 3 |

| REMARKS: | |
|---|---|
| 1 | ASME RATED. |
| 2 | FACTORY PRE-CHARGE PRESSURE 12 PSIG. CONTRACTORS TO FINALIZE REQUIRED CHARGE PRESSURE AT SYSTEM STARTUP. |
| 3 | PROVIDE WITH REPLACEABLE ACCEPTANCE BLADDER DESIGN. |

## BUFFER TANK SCHEDULE

| TAG | TYPE | SERVICES | FLUID | MIN. STORAGE VOLUME (GALLONS) | PIPE CONNECTIONS (INCH) | MANUFACTURER | MODEL NUMBER | REMARKS |
|---|---|---|---|---|---|---|---|---|
| BT-1 | VERTICAL | CHILLED WATER SYSTEM (RETURN SIDE) | 30% GLYCOL | 130 | 2" | NILES | CBT-24-072 | 1, 2, 3, 4 |

| REMARKS: | |
|---|---|
| 1. | TANK TO BE ASME RATED, PROVIDE ASME RATED T&P RELIEF |
| 2. | PROVIDE ALUMINUM CLAD JACKETING & INSULATION. |
| 3. | COORDINATE EQUIPMENT DIMENSIONS WITH MECHANICAL ROOM DOOR AND CONNECTIONS WITH SCHEMATICS. |
| 4. | DESIGN BASIS TANK OVERALL DIAMETER 24-INCH (EXCLUDING FLANGES), HEIGHT 72-INCH, AND A DRY WEIGHT OF 380LBS / OPERATIONAL WEIGHT OF 1,700 LBS. |
| 5. | PROVIDE 4" CONCRETE HOUSE KEEPING PAD, PAD SHALL EXTEND 4" BEYOND FOOTPRINT OF EQUIPMENT. |
| 6. | PROVIDE WITH MANWAY. |

STAMP

COLORADO LICENSED
46310
PROFESSIONAL ENGINEER
9/22/2026


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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLORADO RD,
GOLDEN, CO 80401

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| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME
MECHANICAL SCHEDULES

SHEET NO.
M-700


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## GENERAL REQUIREMENTS:

1. ALL WORK, MATERIALS, AND TRADES SHALL FOLLOW THE LATEST ADOPTED EDITIONS OF THE NATIONAL FIRE PROTECTION ADMINISTRATION (NFPA) AND INTERNATIONAL CODE COUNCIL (ICC) STANDARDS FOR BUILDING CONSTRUCTION AND ENERGY USE. REFER TO ARCHITECTURAL CODE SHEET FOR ADDITIONAL INFORMATION. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PERFORM WORK IN CONFORMANCE WITH ALL APPLICABLE CODES, ORDINANCES, AND LIFE SAFETY MEASURES AS REQUIRED BY LOCAL, STATE, AND NATIONAL AUTHORITIES. CONSULT LITERATURE AND WEBSITES OF JURISDICTIONAL AUTHORITIES FOR ANY AMENDMENTS OR ADDITIONAL REQUIREMENTS.
2. ALL MATERIALS SHALL BE NEW AND LISTED FOR THE APPLICATION UNLESS OTHERWISE NOTED. EQUIPMENT SHALL BE SET STRAIGHT AND SQUARE WITH THE STRUCTURE. CONDUIT SHALL BE INSTALLED PARALLEL OR AT RIGHT ANGLES TO ALL WALLS AND STRUCTURAL MEMBERS. HANGER RODS, EQUIPMENT BASES, SUPPORTS, AND ISOLATION SHALL BE SET PLUMB AND SQUARE AND IN VERTICAL ALIGNMENT WHERE APPLICABLE. ALL WORK SHALL BE NEAT AND CLEAN UPON COMPLETION. ALL WORK SHALL BE PERFORMED IN A WORKMANLIKE MANNER TO THE SATISFACTION OF THE OWNER AND ENGINEER.
3. CONTRACTOR IS RESPONSIBLE FOR A COMPLETE AND OPERATIONAL INSTALLATION OF ALL SYSTEMS DEPICTED WITHIN THESE CONTRACT DOCUMENTS, ALONG WITH STARTUP AND COMMISSIONING AS DESCRIBED IN THE SPECIFICATIONS. REASONABLE ASSUMPTIONS SHALL BE MADE FOR ANY ASPECT NOT COMPLETELY DETAILED WITHIN AND ALL LABOR AND MATERIALS CONTAINED IN THE FINAL BID PRICE. THE CONTRACTOR SHALL ASSURE AND DEMONSTRATE PROPER OPERATION OF ALL DESCRIBED SYSTEMS AND EQUIPMENT TO THE OWNER PRIOR AS PART OF SUBSTANTIAL COMPLETION.
4. ALL DRAWINGS AND SPECIFICATIONS ARE TO BE READ IN COORDINATION WITH ONE ANOTHER SUCH THAT WORK SHOWN ON ONE SET IS BINDING TO THE OTHER. WHERE A DISCREPANCY EXISTS WITHIN THESE CONTRACT DOCUMENTS, THE SPECIFICATIONS SHALL TAKE PRECEDENCE. WHERE A DISCREPANCY EXISTS WITHIN THE SPECIFICATION, CONTRACTOR SHALL PRICE THE MORE EXPENSIVE APPROACH; AND THE DETAILS SHALL BE WORKED OUT DURING CONSTRUCTION.
5. THESE DRAWINGS AND SPECIFICATIONS ARE NOT A DETAILED SET OF INSTALLATION INSTRUCTIONS BUT RATHER A GUIDELINE TO ESTABLISH QUALITY OF EQUIPMENT AND MATERIALS. LEVEL OF WORKMANSHIP, AND PERFORMANCE. DO NOT SCALE DRAWINGS. VERIFY DIMENSIONS WITH SHOP DRAWINGS COORDINATION EFFORT AND IN THE FIELD PRIOR TO COMMENCEMENT OF WORK.
6. CONTRACTOR SHALL BE RESPONSIBLE TO REPAIR, UP TO A LIKE-NEW CONDITION, ANY AND ALL DAMAGES CAUSED TO EXISTING EQUIPMENT AND/OR ADJACENT STRUCTURES. ALL REPAIRED SURFACES SHALL BE INDISTINGUISHABLE FROM THE EXISTING ADJACENT SURFACES. PREPARE LISTING OF ALL EXISTING DAMAGED ITEMS AND SUBMIT TO OWNER PRIOR TO BEGINNING WORK. ITEMS NOT DOCUMENTED BECOME A RESPONSIBILITY OF THE CONTRACTOR.
7. THE CONTRACTOR SHALL REMOVE AND PROPERLY DISPOSE OF ALL DEBRIS AND SHALL CLEAN ALL EQUIPMENT AND AREAS AFFECTED BY CONTRACTOR'S WORK AT THE END OF EACH SHIFT EACH DAY. DISPOSAL OF ALL MATERIALS SHALL BE AS DIRECTED BY THE DIVISION 01 SPECIFICATIONS.
8. SYMBOLS, ABBREVIATIONS, AND DEVICES IN THE LEGENDS ARE GENERAL. FOR EXACT REQUIREMENTS; REFER TO THE SCHEDULES, LAYOUTS, DETAILS, AND TO THE SPECIFICATIONS SINCE THE APPEARANCE OF A PARTICULAR ITEM IN THE LEGEND DOES NOT NECESSARILY IMPLY THAT THE ITEM IS INCLUDED AS A REQUIREMENT OF THE DESIGN.
9. ALL PENETRATIONS FOR RACEWAY AND CABLE THAT ARE BY THIS CONTRACTOR AND WHICH PENETRATE FLOORS, FIRE AND/OR SMOKE BARRIERS, AND SEPARATIONS SHALL BE SEALED WITH A SYSTEM SPECIFICALLY U.L. APPROVED FOR THE APPLICATION.
10. ALL NEW FEEDER, HOMERUN, AND BRANCH CIRCUITS SHALL BE INSTALLED WITH A SEPARATE INSULATED GROUND CONDUCTOR. ANY EXISTING RACEWAYS CURRENTLY BEING USED AS GROUNDING PATH AND PART OF THE SCOPE OF THIS PROJECT SHALL BE PROVIDED WITH NEW GROUND WIRE.
11. THIS PROJECT CONTAINS MULTIPLE PHASES OF CONSTRUCTION. SEE SPECIFICATIONS AND ELECTRICAL PHASING SHEETS FOR CONSTRUCTION PHASING REQUIREMENTS. TEMPORARY ELECTRICAL SERVICES WILL BE REQUIRED. DURING EACH PHASE, AS MUCH WORK AS POSSIBLE MUST BE PERFORMED WITHIN THE BOUNDARIES OF THAT PHASE.

## COORDINATION NOTES:

1. THIS SHEET IS PART OF A COMPREHENSIVE SET OF DESIGN DRAWINGS AND SPECIFICATIONS THAT INCLUDE OTHER ENGINEERING, ARCHITECTURAL, AND DESIGN DISCIPLINES. INSTALLING CONTRACTOR SHALL BE RESPONSIBLE TO REVIEW THE FULL SET OF CONTRACT DOCUMENTS; INCLUDING ALL DRAWINGS, SPECIFICATIONS, AND ADDENDA, IN CONJUNCTION WITH THIS DRAWING PACKAGE TO PROVIDE AN EFFICIENTLY SEQUENCED AND FULLY COORDINATED INSTALLATION. PROVIDE COORDINATION DRAWINGS PER SPECIFICATIONS. CONTRACTOR BEARS THE RESPONSIBILITY TO PROVIDE ANY ADDITIONAL LABOR AND MATERIALS THAT RESULT FROM A LACK OF INTERDISCIPLINARY COORDINATION.
2. ITEMS SPECIFIED HEREIN BY MANUFACTURER AND MODEL NUMBER ARE TO BE CONSIDERED THE MINIMUM QUALITY ACCEPTABLE FOR THE SPECIFIED APPLICATION. FOR SUBSTITUTIONS; IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO PROVIDE SUFFICIENT INFORMATION TO THE DESIGN TEAM IN ORDER TO ESTABLISH "OR EQUAL" STATUS AND TO COORDINATE INSTALLATION OF SAME INTO THE AVAILABLE SPACE. INCREASED COSTS TO THE PROJECT ASSOCIATED WITH CONTRACTOR SUBSTITUTIONS SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR. SUBSTITUTIONS FOR CONVENIENCE WILL NOT BE CONSIDERED PER THE GENERAL REQUIREMENTS OF THE CONTRACT.
3. COORDINATE AND SCHEDULE ALL EQUIPMENT SHUTDOWNS AND ELECTRICAL OUTAGES WITH OWNER AND DESIGN TEAM. PROVIDE A MINIMUM SEVEN (7) CALENDAR DAYS' NOTICE. WHERE OUTAGE AFFECTS BUILDING LIFE SAFETY SYSTEMS, CONTRACTOR SHALL PREPARE AND SUBMIT A WRITTEN METHOD OF PROCEDURES.
4. FURNISH ALL NECESSARY SLEEVES, FLASHINGS, COUNTER-FLASHINGS, INSERTS, ANCHOR BOLTS, EQUIPMENT SUPPORTS, ETC. TO INSTALLING CONTRACTOR IN A TIMELY FASHION; WHERE REQUIRED TO COMPLETE THE WORK. THIS SHALL INCLUDE ANY AND ALL CAULKING, PATCHING, CUTTING, ETC. AS REQUIRED THROUGH ANY ROOF, FLOOR, AND/OR WALL SYSTEMS.
5. EQUIPMENT SHALL FIT INTO THE SPACE ALLOTTED/INDICATED AND SHALL ALLOW ADEQUATE CLEARANCE FOR ENTRY, INSTALLATION, REPLACEMENT, SERVICING, AND MAINTENANCE. COORDINATE ALL EQUIPMENT WITH WORK BY OTHER TRADES. INSTALLATION SHALL COMPLY WITH MANUFACTURER'S INSTRUCTIONS AND ALL APPLICABLE CODES. CLEARANCES SHALL COMPLY WITH NEC. ALL ADDITIONAL WORK REQUIRED FOR EQUIPMENT PROPOSED BY THE CONTRACTOR SHALL BE IDENTIFIED ON EQUIPMENT SUBMITTALS FOR REVIEW BY THE EOR PRIOR TO PURCHASE AND SHALL BE PROVIDED AT NO ADDITIONAL COSTS.
6. DO NOT CUT, CORE, AFFIX TO, OR MODIFY ANY STRUCTURAL OR BUILDING ELEMENTS WITHOUT RECEIVING PRIOR APPROVAL FROM THE ENGINEER OR ARCHITECT INVOLVED.
7. THE CONTRACTOR SHALL MAINTAIN AN ACCURATE SET OF RECORD DRAWINGS ON-SITE AT ALL TIMES DURING CONSTRUCTION WHICH SHALL BE AVAILABLE FOR EXAMINATION BY THE OWNER'S REPRESENTATIVE UPON REQUEST. PROVIDE THE OWNER WITH A COMPLETE MARKED-UP SET OF AS-BUILT CONTRACT DRAWINGS AT PROJECT COMPLETION. REFER TO SPECIFICATIONS FOR ADDITIONAL RECORD DRAWING REQUIREMENTS.

## DEMOLITION NOTES:

1. CONTRACTOR SHALL MAKE ARRANGEMENTS WITH OWNER AND/OR ARCHITECT TO VISIT SITE, ALONG WITH SUBCONTRACTORS, PRIOR TO SUBMITTING A COST PROPOSAL TO ENSURE THE TRADES ARE FAMILIAR WITH THE SPECIFIC CONDITIONS GOVERNING THIS PROJECT SCOPE. PAY PARTICULAR ATTENTION TO SCOPE AREAS AS THEY RELATE TO EXISTING SYSTEMS AND ANY PHASING INDICATED. FIELD VERIFY EXACT EXTENT OF DEMOLITION WORK. NO EXTRAS WILL BE CONSIDERED FOR WORK REQUIRED TO ACHIEVE THE INTENT OF THESE CONTRACT DOCUMENTS. NOTIFY DESIGN TEAM OF ANY DISCREPANCIES FOR RESOLUTION AS SOON AS DISCOVERED.
2. EXISTING CONDITIONS SHOWN ON THESE PLANS REPRESENT LIMITED FIELD OBSERVATIONS MADE BY ST+B ENGINEERING. ROUTING, HIDDEN CONDITIONS, AND CONFORMANCE TO CURRENT PREVAILING CODES HAS NOT BEEN VERIFIED. DESIGN TEAM SHALL NOT BE RESPONSIBLE FOR THE ACCURACY AND/OR SAFETY OF CONDITIONS INDICATED AS EXISTING.
3. DEMOLITION ITEMS SHALL BE REMOVED COMPLETELY; INCLUDING ALL CONNECTIONS, MOUNTING, PATHWAYS, SUPPORTS, AND WIRING. CIRCUIT SHALL BE REMOVED BACK TO LAST ACTIVE BRANCH, DEVICE, OR PANEL; UNLESS SPECIFICALLY NOTED FOR REUSE OR TO BE TURNED OVER TO OWNER. CONTRACTOR SHALL HAVE RESPONSIBILITY TO PROPERLY RECYCLE AND/OR DISPOSE OF ALL REMOVED ITEMS. DO NOT BURN OR BURY.
4. EXISTING CONDUITS MAY BE REUSED WHERE THEY COMPLY WITH THE REQUIREMENTS OF THESE SPECIFICATION. WIRING AND PATHWAY EXTENSIONS SHALL BE NEW. DO NOT REINSTALL ANY CONDUIT WHICH IS REMOVED OR DEMOLISHED. PATHWAYS ROUTING WITHIN OR PASSING THROUGH CONCRETE FLOOR SLABS SHALL BE CUT BACK TO JUST BELOW CONCRETE AND FILLED WITH GROUT. FINISH TO ACHIEVE A SMOOTH AND EVEN SURFACE FLUSH WITH SURROUNDING CONCRETE. ALL PATHWAYS, NEW AND EXISTING UNDER THIS CONTRACT, SHALL BE LABELED PER THE SPECIFICATIONS.
5. ALL WALLS SHALL BE PROPERLY PATCHED AND REPAINTED AFTER THE REMOVAL OF DEVICES. REPAIR TO A LEVEL OF FINISH CONSISTENT WITH THE ADJACENT SURFACES. BLANK COVER PLATES OVER EXISTING BOXES ARE NOT ACCEPTABLE, UNLESS NOTED OTHERWISE.
6. RE-SUPPORT AND SECURE EXISTING RACEWAYS/CONDUITS/BUILDING WIRE/LOW VOLTAGE CABLE ASSOCIATED WITH SCOPE OF WORK TO COMPLY WITH CURRENT ADOPTED REQUIREMENTS OF NFPA 70, CHAPTER 3.
7. MAINTAIN CIRCUIT CONTINUITY TO EXISTING CIRCUITS AND DEVICES TO REMAIN. REWORK EXISTING WIRING, BOXES, TERMINATIONS, ETC. AS REQUIRED TO INTERFACE NEW WORK SHOWN.
8. WHERE THE REMOVAL OF A SURFACE MOUNTED RACEWAY/DEVICE LEAVES VISIBLE EVIDENCE ON AN EXISTING SURFACE NOT SCHEDULED FOR DEMOLITION OR REWORK, THE SURFACE SHALL BE REPAIRED OR REPLACED TO A QUALITY EQUAL TO FINAL SCHEDULED FINISH OF ADJACENT SURFACES. WHERE THE EXISTING RACEWAY IS CONCEALED IN ASSEMBLIES NOT SCHEDULED FOR DEMOLITION THE OUTLET BOX SHALL BE CLEANED, AND A BLANK COVER-PLATE INSTALLED.

## POWER NOTES:

1. ALL DEDICATED WORKING SPACES AS DEFINED IN THE NEC SHALL BE MARKED OUT AND FIELD PAINTED ON THE FLOOR BENEATH THE EQUIPMENT. PROVIDE A FOUR-INCH WIDE YELLOW AND BLACK OUTLINE OF THE WORKING SPACE PERIMETER IN PAINT OR PERMANENT VINYL, FLOOR-MARKING TAPE.
2. PANEL SCHEDULES SHALL BE FILLED OUT SO EVERY NEW AND/OR MODIFIED CIRCUIT IS UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.
3. PROVIDE NYLON PULL ROPE FOR ALL SPARE CONDUITS AS WELL AS ANY PATHWAYS PROVIDED FOR OTHER TRADES SUCH AS CONTROLS OR TECHNOLOGY.
4. INSTALL ALL CONDUIT GENERALLY IN THE ROUTES/AREAS SHOWN ON DRAWINGS. PROVIDE ADDITIONAL SUPPORT WHEREVER THE BUILDING STRUCTURE IS NOT SUITABLE FOR DIRECT MOUNTING. OFFSET CONDUIT, EQUIPMENT, ETC. AS NECESSARY TO ACCOMMODATE STRUCTURE, BEAMS, COLUMNS, OTHER SYSTEMS IN THIS TRADE AND OTHER TRADES. CONDUIT OR EQUIPMENT THAT IS NOT INSTALLED PROPERLY OR WITHIN THE SPACE PROVIDED SHALL BE REMOVED AND REINSTALLED AT NO EXTRA COST.

## ONE LINE DIAGRAM NOTES:

1. PANELBOARD AND BREAKER INTERRUPTING CAPACITIES (AIC) SHALL BE FULLY RATED AS AN ASSEMBLY. SERIES RATING OF BREAKER DEVICES AND EQUIPMENT IS NOT PERMITTED.
2. ALL CONDUCTOR SIZES ARE BASED ON COPPER. ALUMINUM IS ALLOWED UNDER SPECIFIC CONDITIONS. SEE SPECIFICATIONS FOR ACCEPTABLE USE. WHERE ALUMINUM IS SUBSTITUTED, RACEWAY FILL CALCULATIONS SHALL BE A RESPONSIBILITY OF THE CONTRACTOR.
3. WHERE SPECIFIED WIRE SIZE IS NOT A STOCKED ITEM AND INCURS COSTS ABOVE THE NEXT LARGER TRADE SIZE, IT SHALL BE ACCEPTABLE TO SUBSTITUTE WITH THE MORE COST-EFFECTIVE WIRE SIZE. CONDUIT SIZING AND GROUND CONDUCTOR SHALL BE ADJUSTED ACCORDINGLY AS A RESPONSIBILITY OF THE CONTRACTOR.
4. VERIFY MECHANICAL EQUIPMENT AND CONTROLLERS ARE RATED FOR AVAILABLE SHORT CIRCUIT CURRENTS. PROVIDE FUSED DISCONNECT WITH CURRENT LIMITING RK1 TYPE FUSES AS REQUIRED.
5. DO NOT SPLICE FEEDER CONDUCTORS TO PANELBOARDS, SWITCHBOARDS, MOTOR CONTROL CENTERS, OR TO MOTORS UNLESS PRIOR APPROVAL HAS BEEN OBTAINED FROM THE ENGINEER.
6. PROVIDE SEPARATE INSULATED GROUNDING CONDUCTOR IN ALL NEW FEEDER, HOMERUN AND BRANCH CIRCUITS. ANY EXISTING RACEWAYS BEING USED AS GROUNDING PATH, AND PART OF THE SCOPE OF THIS PROJECT, SHALL BE PROVIDED WITH NEW GROUND WIRE.
7. ELECTRICAL CONTRACTOR TO TEST VOLTAGES AT TRANSFORMER SECONDARIES AND TAP TRANSFORMERS UP AS NECESSARY TO ENSURE EQUIPMENT RATED VOLTAGE IS PROVIDED AT TRANSFORMER SECONDARY.
8. ONE-LINE DIAGRAMS ARE ONLY INTENDED TO DEPICT FEEDER CIRCUITS. BRANCH CIRCUITS ARE NOT GENERALLY CALLED OUT. REFER TO PANELBOARD SCHEDULE(S) FOR REQUIREMENTS. ALL 20A/1P BRANCH CIRCUITS SHALL BE WIRED WITH #12 AWG CONDUCTORS AT A MINIMUM UNLESS OTHERWISE NOTED.
9. UPDATE PANEL SCHEDULE TO REFLECT CIRCUIT ARRANGEMENTS AFTER CONTRACTOR HAS BALANCED LOAD TO WITHIN 20%. NEW CIRCUIT DIRECTORIES SHALL BE TYPEWRITTEN.
10. NEW BREAKERS SHALL BE COMPATIBLE WITH AND LISTED FOR USE IN EXISTING EQUIPMENT.

## EQUIPMENT CONNECTION NOTES:

1. MECHANICAL EQUIPMENT DENOTED ON PLANS BY EQUIPMENT TAGS. REFER TO MECHANICAL EQUIPMENT SCHEDULES FOR ADDITIONAL INFORMATION. COORDINATE EXACT CONNECTION REQUIREMENTS, OUTLET CONFIGURATIONS, AND/OR CONDUIT ENTRY LOCATIONS OF MECHANICAL EQUIPMENT WITH INSTALLING CONTRACTOR. NO EXTRAS WILL BE ALLOWED FOR ADDITIONAL CIRCUITS AND ACCESSORY CONNECTIONS SHOWN ON APPROVED MECHANICAL SHOP DRAWINGS. REFER TO SPECIFICATIONS FOR COORDINATION DRAWING REQUIREMENTS.
2. MAKE ALL FINAL ELECTRICAL TERMINATIONS TO EQUIPMENT REQUIRING POWER AND/OR CONTROL, THIS SHALL INCLUDE BUT NOT BE LIMITED TO ALL: MECHANICAL, OWNER, SPECIALTY, AND/OR FF&E INCLUDED IN THIS PROJECT. COORDINATE EXACT CONNECTION REQUIREMENTS, OUTLET CONFIGURATIONS, AND/OR CONDUIT ENTRY LOCATIONS OF EQUIPMENT WITH INSTALLING CONTRACTOR.
3. PIPING AND / OR DUCTWORK SHALL NOT BE ROUTED IN THE DEDICATED ELECTRICAL EQUIPMENT SPACE OF ELECTRICAL SWITCHBOARDS, DISTRIBUTION BOARDS, PANELBOARDS, ETC. PROVIDE DEPTH AND WIDTH OF WORKING SPACE FOR ELECTRICAL EQUIPMENT AS REQUIRED BY NEC.
4. BUILDING POWER HAS BEEN SHOWN ON DRAWINGS TO J-BOXES FOR ANCILLARY POWER SUCH AS BAS/BMS CONTROLLER, DAMPER MOTORS, ETC. COORDINATE FINAL REQUIREMENTS WITH MECHANICAL CONTROLS CONTRACTOR AND INSTALLED EQUIPMENT. CIRCUITS SHALL BE DEDICATED TO THAT PURPOSE AND SHALL NOT CONTAIN ANY OTHER "GENERAL" LOADS.
5. PROVIDE ADDITIONAL SUPPORT FOR SWITCHES, STARTERS, RACEWAYS, AND OTHER ELECTRICAL EQUIPMENT WHEREVER THE BUILDING STRUCTURE IS NOT SUITABLE FOR DIRECT MOUNTING.
6. PROVIDE DEEP, 4-SQUARE, ELECTRICAL BOX, SINGLE GANG MUD RING AND BLANK SINGLE GANG COVER PLATE FOR ALL LINE VOLTAGE THERMOSTATS. ROUTE 1/2" CONDUIT TO EQUIPMENT. THERMOSTAT DEVICE IS FURNISHED WITH EQUIPMENT OR BY OTHERS. INSTALL AND MAKE FINAL WIRE TERMINATIONS.
7. LOCATE ALL OUTLETS AND DISCONNECT SWITCHES WHERE ACCESSIBLE. PROVIDE WORKING CLEARANCE AS REQUIRED BY NEC. PROVIDE ACCESS HATCH WHERE LOCATED ABOVE HARD LID CEILING.
8. VAV UNITS SHOWN WITH INTEGRAL 120V/24V TRANSFORMER. IF A CENTRAL POWER SUPPLY IS FURNISHED, PROVIDE CONNECTIONS TO EACH UNIT AS REQUIRED.
9. REFER TO SPECIALTY DRAWING PACKAGES FOR POOL/KITCHEN/SPA/ARCADE/LAB/ELEVATOR EQUIPMENT MODEL NUMBERS, DIMENSIONS, ANTICIPATED DIVISION OF TRADES, AND SPECIFIC ELECTRICAL REQUIREMENTS, OPTIONS, AND ACCESSORIES. INFORMATION ON SPECIALTY DRAWING PACKAGE SUPERSEDES INFORMATION SHOWN ON ELECTRICAL AND/OR TECHNOLOGY DRAWINGS. SHOULD THERE BE ANY DISCREPANCY BETWEEN ACTUAL EQUIPMENT INSTALLED AND THESE DRAWINGS; CONTRACTOR SHALL NOTIFY ENGINEER IMMEDIATELY FOR CLARIFICATION AND FURTHER DIRECTION. VENDOR'S INSTALLATION REQUIREMENTS ARE MADE PART OF CONTRACT DOCUMENTS BY THIS REFERENCE.
10. REFERENCE PHILIPS HEALTHCARE, SITE PREPARATION DOCUMENT #N-WES170614-H (DATED 10/10/18) FOR DIMENSIONED LOCATIONS AND ADDITIONAL INFORMATION REGARDING INSTALLATION REQUIREMENTS OF BI-PLANE, CT AND SUPPORT EQUIPMENT. VENDOR'S INSTALLATION REQUIREMENTS ARE MADE PART OF CONTRACT DOCUMENTS BY THIS REFERENCE.
11. PROVIDE RACEWAYS AND PULL STRINGS AS REQUIRED TO SUPPORT WIRING ASSOCIATED WITH GE EQUIPMENT. REFER TO EQUIPMENT COORDINATION DRAWINGS FROM GE (DATED 11/30/20) FOR CONDUIT SIZES, QUANTITIES AND END POINTS.

## ELECTRICAL ONE-LINE LEGEND

| PLAN | DESCRIPTION |
|---|---|
| | CIRCUIT BREAKER |
| | DISCONNECT SWITCH |
| M | UTILITY METER |
| | CURRENT TRANSFORMER |
| | GROUND |
| | CONTACTOR |
| C | COIL |
| | SERVICE WEATHERHEAD |
| M | MOTOR |
| | DISCONNECT, SOLID NEUTRAL, FUSED |
| | DISCONNECT, SOLID NEUTRAL, NON-FUSED |
| | MOTOR STARTER / CONTACTOR |
| | FUSE |
| | ENCLOSED CIRCUIT BREAKER |
| J | JUNCTION BOX |
| A | AMP METER |
| V | VOLT METER |
| K | KIRK KEY INTERLOCK SYSTEM |
| GFP | GROUND FAULT PROTECTION |
| ST | SHUNT TRIP |
| | FAULT CURRENT |
| VFD | VARIABLE FREQUNCY DRIVE |
| | EMERGENCY POWER OFF |
| SPD | SURGE PROTECTION DEVICE |
| PANEL XXXX | ELECTRICAL PANELBOARD |
| E N L | AUTOMATIC TRANSFER SWITCH |
| T ## | STEP DOWN TRANSFORMER (## INDICATES KVA) |
| TK ## | K-RATED STEP DOWN TRANSFORMER (## INDICATES KVA) |
| | SERVICE ENTRANCE TRANSFORMER |
| E G R L | GENERATOR |
| RAP | GENERATOR REMOTE ANNCIATOR PANEL |

## CABLE / PATHWAY LEGEND

| PLAN | DESCRIPTION |
|---|---|
| | CEILING SPACE / OVERHEAD WIRING |
| | UNDERGROUND / UNDER SLAB WIRING |
| | EXPOSED PATHWAY |
| | CONTROLS AND/OR SIGNAL WIRING |
| WM | SURFACE MOUNT PATHWAY (WIREMOLD) |
| | PATHWAY / CABLE TURNS UP, TURNS DOWN |

## POWER LEGEND

| ELECTRICAL DEVICES CLG. | FLR. | WALL | DESCRIPTION |
|---|---|---|---|
| | | | SINGLE RECEPTACLE |
| | | | DUPLEX RECEPTACLE |
| | | | GFI DUPLEX RECEPTACLE |
| | | | QUAD RECEPTACLE |
| | | | GFI QUAD RECEPTACLE |
| | | | SPECIAL RECEPTACLE (SUBSCRIPT DENOTES TYPE) |
| | | | JUNCTION BOX |
| | | | HALF SWITCHED RECEPTACLE |
| | | | ISOLATED GROUND RECEPTACLE |
| - | - | | SINGLE POLE SWITCH |
| - | - | SU | FUSED SWITCH |
| - | - | M | MANUAL SWITCH, HORSEPOWER RATED |
| - | - | P | SWITCH WITH PILOT LIGHT |
| - | - | TO | THERMAL OVERLOAD SWITCH |
| - | - | | PANELBOARD |
| - | - | | GROUND BAR |
| - | - | | DISCONNECT, SOLID NEUTRAL, FUSED |
| - | - | | DISCONNECT, SOLID NEUTRAL, NON-FUSED |

PLAN CALLOUT:

| | | |
|---|---|---|
| DEVICE MOUNTED ABOVE COUNTER | = | AC |
| DEVICE MOUNTED UNDER COUNTER | = | UC |
| DEVICE MOUNTED AT SPECIFIC ELEVATION | = | +48" |
| HARDENEDWEATHERPROOF DEVICE & IN USE COVER | = | WP |
| DUPLEX WITH TWO INTEGRAL USB PORTS | = | USB-C |

## FIRE ALARM LEGEND

| PLAN | DESCRIPTION |
|---|---|
| H F | AUDIBLE SIGNALING DEVICE (H = HORN, S = SPEAKER, C = CHIME) |
| 15cd F | VISUAL SIGNALING DEVICE (cd = CANDELA RATING) |
| H15cd F | AUDIBLE/VISUAL SIGNALING DEVICE (H#cd = HORN/STROBE COMBO, S = SPEAKER/STROBE COMBO, C = CHIME/STROBE COMBO) |
| F | FIRE ALARM PULL STATION |
| | MAGNETIC DOOR HOLD DEVICE |
| SD P | SMOKE DETECTOR (P = PHOTOELECTRIC, SB = w/ SOUNDER BASE, BR = BEAM RECEIVER, BT = BEAM TRANSMITTER) |
| HD F | HEAT DETECTOR (F = FIXED TEMPERATURE, R = FIXED TEMPERATURE & RATE OF RISE (TEMP RATING)) |
| FACP | FIRE ALARM CONTROL PANEL |
| FARA | FIRE ALARM REMOTE ANNUNCIATOR |
| SD | DUCT SMOKE DETECTOR |
| GZM | GRAPHIC ZONE MAP |
| FADD | FIRE ALARM DEVICE DIRECTORY |
| | REMOTE TEST STATION |
| IM | ADDRESSABLE INPUT MODULE |
| OM | ADDRESSABLE OUTPUT MODULE |
| CO | CARBON MONOXIDE DETECTOR |
| TS | TAMPER SWITCH |
| PS | PRESSURE SWITCH |
| FS | FLOW SWITCH |

## ELECTRICAL ABBREVIATIONS

| | |
|---|---|
| A | AMPERE |
| AC | ABOVE COUNTER |
| ADA | AMERICANS WITH DISABILITIES ACT |
| AF/AT | AMP FRAME, AMP TRIP |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| AIC | AMPERES INTERRUPTING CAPACITY (SYMMETRICAL) |
| AS/AF | AMP SWITCH, AMP FUSE |
| ATS | AUTOMATIC TRANSFER SWITCH |
| AWG | AMERICAN WIRE GAUGE |
| BAS | BUILDING AUTOMATION SYSTEM |
| BFG | BELOW FINISHED GRADE |
| C | CONDUIT |
| CB | CIRCUIT BREAKER |
| CKT | CIRCUIT |
| CLG | CEILING |
| CT | CURRENT TRANSFORMER |
| DISC | DISCONNECT |
| (E) | EXISTING |
| EM | EMERGENCY LIGHT WITH UL RELAY FOR SWITCHING |
| EPO / ESD | EMERGENCY POWER OFF / EMERGENCY SHUT DOWN |
| FLA | FULL LOAD AMPS |
| FSD | FIRE SMOKE DAMPER |
| GFCI | GROUND FAULT CIRCUIT INTERRUPTER |
| GFP | GROUND FAULT PROTECTION |
| GND | GROUND |
| HOA | HAND-OFF AUTO SELECTOR |
| HP | HORSEPOWER |
| IG | ISOLATED GROUND |
| KVA | KILOVOLT AMPERES |
| KW | KILOWATT |
| MCA | MINIMUM CIRCUIT AMPACITY |
| MCB | MAIN CIRCUIT BREAKER |
| MCM | THOUSAND CIRCULAR MILS |
| MCP | MOTOR CIRCUIT PROTECTOR |
| MFR | MANUFACTURER |
| MLO | MAIN LUGS ONLY |
| MOCP | MAXIMUM OVERCURRENT PROTECTION |
| NC / NO | NORMALLY CLOSED / NORMALLY OPEN |
| NEC | NATIONAL ELECTRICAL CODE |
| NEMA | NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION |
| NF | NON-FUSED |
| NIC | NOT IN CONTRACT |
| NL | NIGHT LIGHT (SECURITY) |
| NTS | NOT TO SCALE |
| NX | NO CONNECTION |
| OFCI | OWNER FURNISHED, CONTRACTOR INSTALLED |
| PROVIDE | FURNISH AND INSTALL |
| (R) | RELOCATED |
| SPD | SURGE PROTECTION DEVICE |
| TYP | TYPICAL |
| UC | UNDER COUNTER |
| UG | UNDERGROUND |
| UON | UNLESS OTHERWISE NOTED |
| UPS | UNITERRUPTABLE POWER SYSTEM |
| V | VOLTS |
| VA | VOLT AMPERES |
| VFD | VARIABLE FREQUENCY DRIVE |
| WP | WEATHERPROOF |
| XFMR | TRANSFORMER |
| %Z | PERCENTAGE IMPEDANCE |

## PLAN SYMBOLS LEGEND

| PLAN | DESCRIPTION |
|---|---|
| # | DRAWING FLAG NOTE |
| # | DEMOLITION DRAWING FLAG NOTE |
| # | REVISION CLOUD TAG |
| XXX ### | MECHANICAL EQUIPMENT CALLOUT |
| XXX | LIGHTING CONTROL |
| XXX-## | TYPICAL CIRCUIT NUMBER |
| T# | TYPICAL LUMINAIRE TYPE |
| XXX | MISC./OWNER EQUIPMENT CALLOUT |
| XXX ### | MEDICAL EQUIPMENT CALLOUT |
| ### | DOOR NUMBER |
| | ELEVATION/SECTION CALLOUT |

STAMP

COLORADO LICENSED FOOD SPACER 46310 PROFESSIONAL ENGINEER

09/22/2026


ST+B ENGINEERING
2518B GENESEE TRAIL RD, SUITE 120, GOLDEN, CO 80401
(303) 900-8782
HTTP://STB.ENGINEERING


JEFFERSON COUNTY COLORADO
JEFFCO.US (303) 271-6511
100 JEFFERSON CONTY PKWY GOLDEN, CO 80419

# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLOROW RD, GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

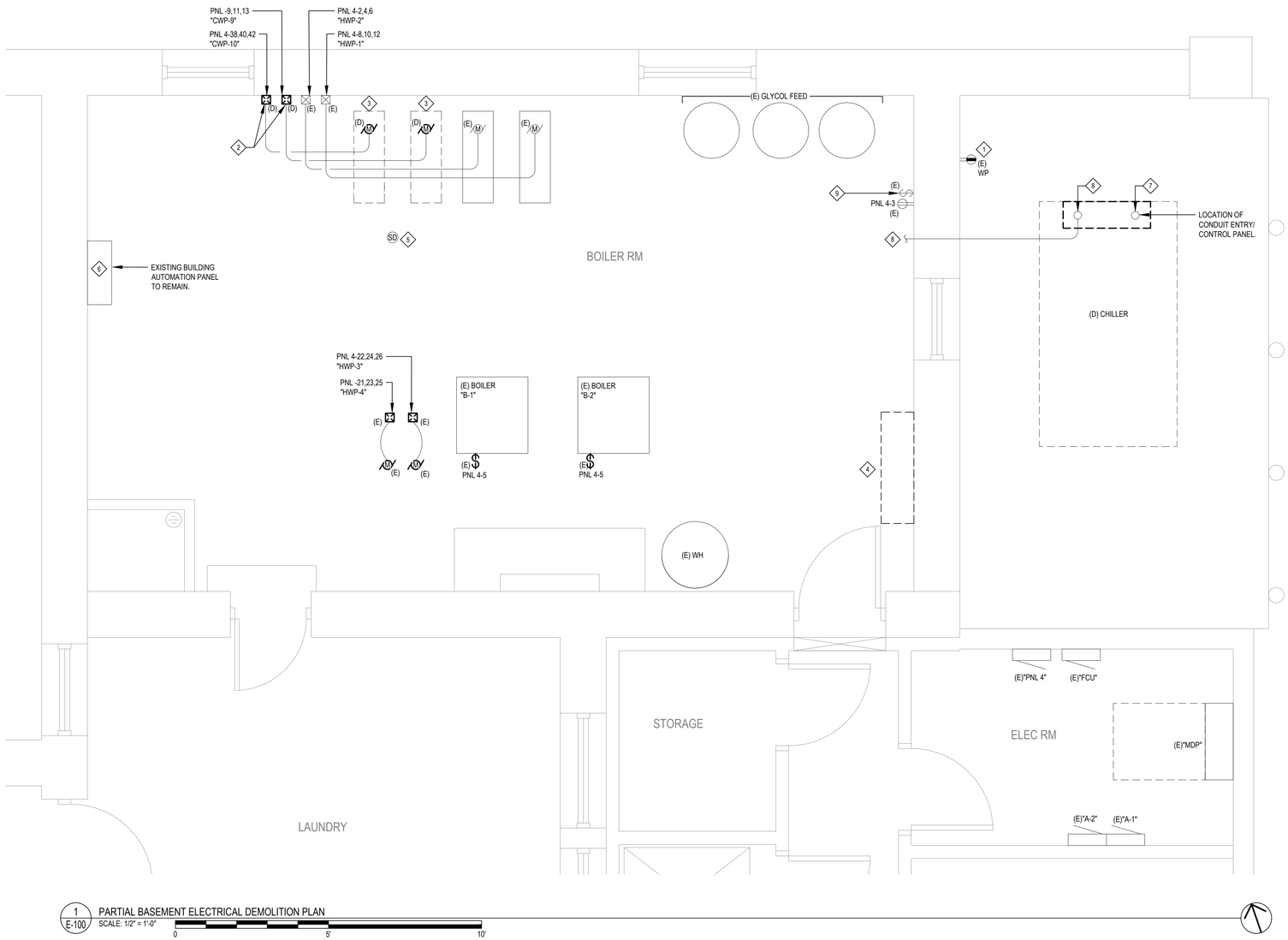
SHEET NAME

ELECTRICAL LEGENDS

SHEET NO.

E-000


24x36 Z:\BOX\STB PROJECTS\2864.18 - JEFFCO - BOETTCHER MANSION EQUIPMENT REPLACEMENT\GOD\DRAWINGS\CHILLER\EOD_LEGEND.DWG
9/22/2026 10:08 AM




1 E-100 PARTIAL BASEMENT ELECTRICAL DEMOLITION PLAN
SCALE: 1/2" = 1'-0"
0 5' 10'

## DEMOLITION KEYNOTES:

1. REFER TO POWER PLAN (NEW WORK) FOR SCOPE PERTAINING TO EXISTING GFCI PROTECTED RECEPTACLE.
2. DISCONNECT AND REMOVE EXISTING MOTOR STARTERS. PROTECT EXISTING BRANCH CIRCUITING FOR REUSE.
3. REMOVE EXISTING CHILLED WATER PUMP. PROTECT EXISTING BRANCH CIRCUITING FOR REUSE.
4. REMOTE FIRE ALARM ANNUNCIATOR PANEL AND TELEPHONE EQUIPMENT AT THIS LOCATION. PROTECT ALL EQUIPMENT DURING DEMOLITION AND CONSTRUCTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DAMAGES TO EXISTING EQUIPMENT DURING CONSTRUCTION. VERIFY AND DOCUMENT WORKING CONDITION OF EQUIPMENT PRIOR TO START OF WORK.
5. PROTECT EXISTING FIRE ALARM SMOKE DETECTOR FROM DUST AND DEBRIS DURING DEMOLITION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR REPLACEMENT SHOULD THE EXISTING DETECTOR BE DAMAGED DURING CONSTRUCTION. PROVIDE DUST COVER TO PREVENT FALSE ALARM. REMOVE DUST COVER AFTER EACH SHIFT OF WORK TO ALLOW PROTECTION. THE CONTRACTOR IS RESPONSIBLE FOR PROVIDING FIRE WATCH WHEN DETECTOR COVER IS IN PLACE.
6. PROTECT ALL EXISTING RACEWAY CONNECTED TO EXISTING BUILDING AUTOMATION PANEL FROM CHILLER AND PUMPS. RECONNECT TO NEW EQUIPMENT AS REQUIRED.
7. APPROXIMATE LOCATION OF CHILLER FEED FROM BELOW GRADE. REMOVE EXISTING CONDUCTORS AND LIQUID TIGHT RACEWAY. PROTECT EXISTING IN-GRADE RACEWAY FOR INSTALLATION OF NEW LIQUID TIGHT RACEWAY AND NEW CONDUCTORS AS NOTED ON POWER PLAN (NEW WORK).
8. PROTECT EXISTING CONTROL CONDUIT ROUTING FROM CHILLER CONTROL PANEL THROUGH WALL. REMOVE EXISTING CABLING. REFER TO POWER PLAN (NEW WORK) FOR ADDITIONAL REQUIREMENTS.
9. DISCONNECT AND REMOVE EXISTING HEAT TRACE CABLING AND CABLE CONNECTOR. PROTECT EXISTING TOGGLE DISCONNECT AND BRANCH CIRCUIT FOR REUSE.

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COLORADO LICENSED
TODD SPACER
46310
PROFESSIONAL ENGINEER
09/22/2026


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100 JEFFERSON CONTY PKWY
GOLDEN, CO 80419

# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLORADO RD,
GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME
PARTIAL BASEMENT ELECTRICAL DEMOLITION PLAN

SHEET NO.
E-100


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## EQUIPMENT SCHEDULE

| EQUIP. ID | EQUIPMENT DESCRIPTION | VOLTS (V) | PH | HP | MCA (A) | FLA (A) | KVA | KW | PANEL | OVERCURRENT PROTECTION | | | FEEDER / CONDUIT | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | DISCONNECT | FUSE | BREAKER | | |
| ACC-1 | AIR COOLED CHILLER | 208 | 3 | - - | 143 1/2 | 114.8 | 41.4 | 51.7 | - - | INTEGRAL | - - | 175 | EXISTING | - - |
| CWP-9 | CHILLED WATER PUMP | 208 | 3 | 5 | - - | 16.7 | 6.0 | 6.0 | - - | VFD | - - | 35 | EXISTING | - - |
| CWP-10 | CHILLED WATER PUMP | 208 | 3 | 5 | - - | 16.7 | 6.0 | 6.0 | - - | VFD | - - | 35 | EXISTING | - - |

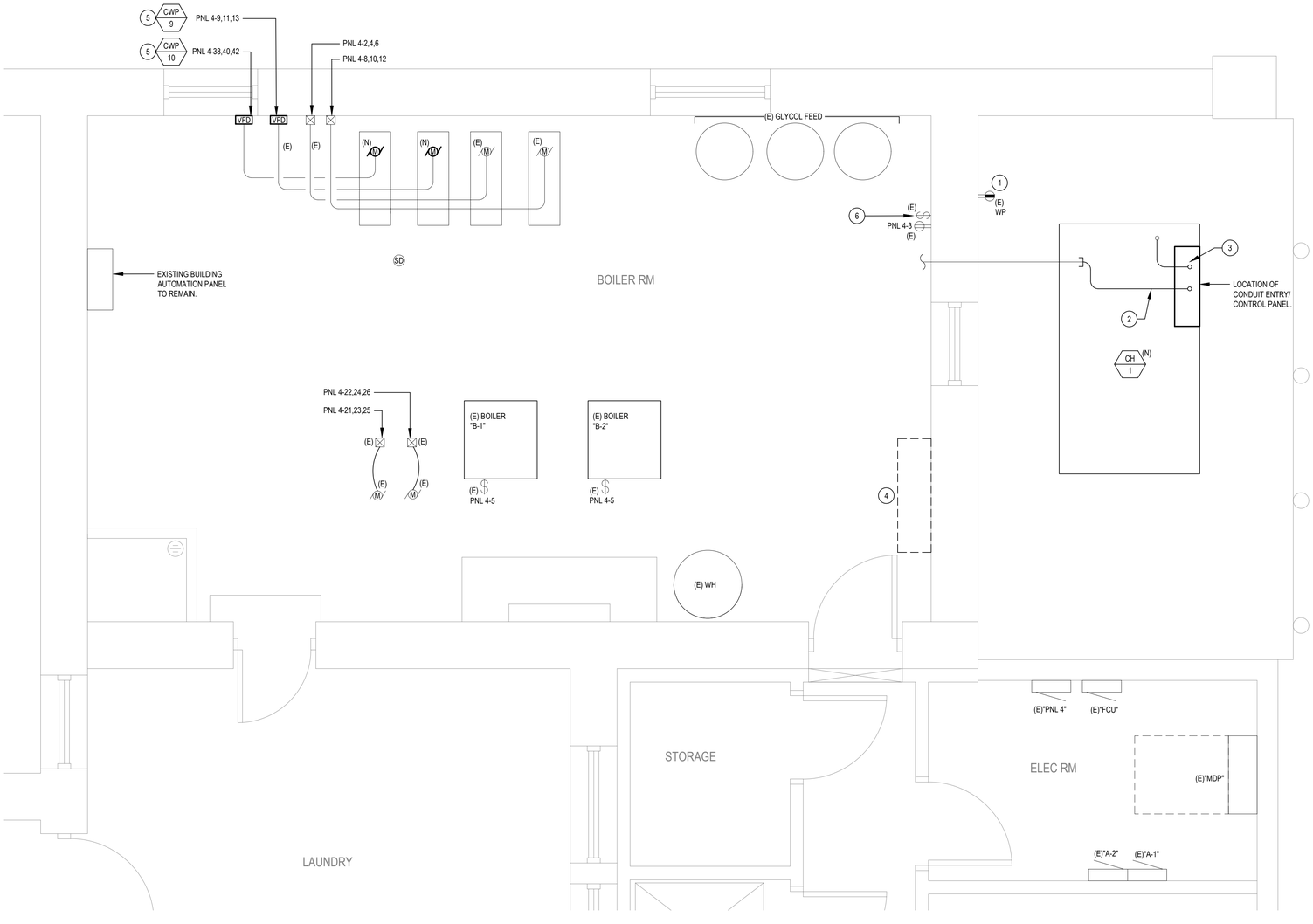
**GENERAL NOTES:**

A. PRIOR TO WORK, CONTRACTOR SHALL VERIFY ELECTRICAL REQUIREMENTS AND DISCONNECTING MEANS FOR EACH PIECE OF EQUIPMENT PER MANUFACTURER'S RECOMMENDATIONS.

B. PRIOR TO WORK, CONTRACTOR SHALL FIELD VERIFY INSTALLATION LOCATION FOR EACH PIECE OF EQUIPMENT WITH OTHER DISCIPLINES PRIOR TO ROUGH-IN.

**SPECIFIC NOTES:**

1. ONE-FOR-ONE EQUIPMENT REPLACEMENT. REUSE EXISTING BRANCH CIRCUITING.




1 E-200 PARTIAL BASEMENT ELECTRICAL NEW WORK PLAN
SCALE: 1/2" = 1'-0"
0 5' 10'

### GENERAL NOTES:

1. REFER TO ELECTRICAL SPECIFICATIONS FOR GRC CONDUIT REQUIREMENTS.

### KEYNOTES:

1. REMOVE EXISTING WEATHER PROOF ACRYLIC COVER AND PROVIDE NEW CAST ALUMINUM "IN-USE" COVER, HUBBELL #WP26E (OR EQUIVALENT).
2. EXTEND 3/4" EMT FROM EXISTING EMT LOCATION TO NEW UNIT CONTROL PANEL.
3. PROVIDE NEW LIQUID TIGHT FLEXIBLE RACEWAY FROM EXISTING CONDUIT STUB UP LOCATION TO NEW MAIN POWER CONNECTION LOCATION IN UNIT CONTROL PANEL. PROVIDE NEW CONDUCTORS FROM CHILLER TO EXISTING CIRCUIT BREAKER IN MDP. REFER TO ELECTRICAL ONE-LINE DIAGRAM (NEW WORK) FOR CONDUCTOR REQUIREMENTS. SPLICES SHALL NOT BE PERMITTED.
4. REMOTE FIRE ALARM ANNUNCIATOR PANEL AND TELEPHONE EQUIPMENT AT THIS LOCATION. PROTECT ALL EQUIPMENT DURING DEMOLITION AND CONSTRUCTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DAMAGES TO EXISTING EQUIPMENT DURING CONSTRUCTION. VERIFY AND DOCUMENT WORKING CONDITION OF EQUIPMENT PRIOR TO START OF WORK.
5. CONNECT NEW VFD TO EXISTING BRANCH CIRCUITING PROTECTED DURING DEMOLITION. REUSE EXISTING BRANCH CIRCUITING BETWEEN VFD AND MOTOR. INSTALL VFD IN SAME LOCATION AS STARTER REMOVED DURING DEMOLITION. MAKE FINAL CONNECTIONS TO NEW PUMP MOTOR.
6. PROVIDE NEW RAYCHEM #RAYCLIC-PC AND APPROXIMATELY 24' OF HEAT CABLE. CONNECT TO EXISTING BRANCH CIRCUIT/TOGGLE DISCONNECT PROTECTED DURING DEMOLITION. HEAT TRACE SHALL BE REINSTALLED ON SUPPLY AND RETURN CHILLER LINES MATCHING ORIGINAL INSTALLATION.

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# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLOROW RD,
GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME

PARTIAL BASEMENT ELECTRICAL NEW WORK PLAN

SHEET NO.

E-200


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## SWITCHBOARD: MDP (EXISTING)

| | |
|---|---|
| **FED FROM:** | POLE MOUNTED UTILITY TRANSFORMER |
| **LOCATION:** | SEE PLANS |
| **MOUNTING:** | SURFACE |
| **ENCLOSURE:** | NEMA 1 |

| | |
|---|---|
| **VOLTAGE:** | 208Y/120V, 3PH, 4W |
| **BUS AMPACITY:** | 800A |
| **MAIN:** | MCB |
| **AIC:** | 22,000A |

| CKT NO | LOAD DESCRIPTION | LOAD VA | LOAD TYPE | FRAME SIZE | TRIP RATING | # OF POLES | NOTES | PHASE |
|---|---|---|---|---|---|---|---|---|
| 1 | PANEL "C" | 72060 | R | 400A | - - | 3 | | A |
| 2 | PANEL "A" SECTION 1 AND 2 | 112500 | E | 225A | - - | 3 | | B |
| 3 | PANEL "B" | 36030 | R | 200A | - - | 3 | | C |
| **4** | **CHILLER** | **39777** | **M** | **175A** | **- -** | **3** | **1** | **A** |
| **5** | **PANEL "PNL 4"** | **41803** | **0** | **150A** | **- -** | **3** | **1** | **B** |
| 6 | OUTSIDE LIGHTING/POLE LIGHT BY DUMPSTER | 2500 | L | 20A | - - | 2 | | C |
| 7 | SPARE | | | 100A | - - | 3 | | A |
| 8 | POLE LIGHTS | 2500 | L | 40A | - - | 1 | | B |
| 9 | 3-POLE SPACE | | | | | 3 | | C |
| 10 | 3-POLE SPACE | | | | | 3 | | A |

| LOAD TYPES | CONNECTED LOAD (VA) A | B | C |
|---|---|---|---|
| LIGHTING | 0 | 2500 | 2500 |
| RECEPTACLES | 72060 | 0 | 36030 |
| MOTORS | 39777 | 0 | 0 |
| LARGEST MOTOR | 0 | 0 | 0 |
| EQUIPMENT | 0 | 112500 | 0 |
| APPLIANCES | 0 | 0 | 0 |

| SUBFEED TOTAL | SWITCHBOARD TOTAL | DEMAND LOAD (VA) FACTOR | TOTAL |
|---|---|---|---|
| 0 | 5000 | 125% | 6250 |
| 0 | 108090 | 55% | 59045 |
| 0 | 39777 | 100% | 39777 |
| 0 | 0 | 125% | 0 |
| 0 | 112500 | 100% | 112500 |
| 0 | 0 | 100% | 0 |
| | **TOTAL DEMAND LOAD (KVA) =** | | **257.3** |
| | **TOTAL DEMAND LOAD (A) =** | | **714** |

**GENERAL NOTES:**

A. EVERY CIRCUIT AND CIRCUIT MODIFICATION SHALL BE UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.

B. PROVIDE NEW UPDATED PANEL DIRECTORIES FOR PANELS THAT ARE INCLUDED IN THIS PROJECT.

C. INDICATE SPARES IN PENCIL.

**SPECIFIC NOTES:**

1. LOAD AFFECTED THIS CONTRACT WITH EQUIPMENT REPLACEMENT. LOAD ADDED IS EQUAL OR LESS THAN LOAD REMOVED. NO NEW LOAD ADDED THIS CONTRACT.

## PANEL: PNL 4 (EXISTING)

| | |
|---|---|
| **FED FROM:** | MDP |
| **LOCATION:** | SEE PLANS |
| **MOUNTING:** | SURFACE |
| **ENCLOSURE:** | NEMA 1 |

| | |
|---|---|
| **VOLTAGE:** | 208Y/120V, 3PH, 4W |
| **BUS AMPACITY:** | 250A |
| **MAIN:** | MLO |
| **AIC:** | 10,000A |

| CKT NO | NOTES | DESCRIPTION | LOAD VA | LOAD TYPE | BREAKER POLE | BREAKER AMPS | PHASE | BREAKER AMPS | BREAKER POLE | LOAD TYPE | LOAD VA | DESCRIPTION | NOTES | CKT NO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | GLYCOL PUMPS | 545 | M | 1 | 20 | A | | | M | 1105 | | | 2 |
| 3 | | HEAT TRACE | 1000 | E | 1 | 20 | B | 35 | 3 | M | 1105 | HEATING WATER PUMP | | 4 |
| 5 | | BOILER CONTROLS | 528 | E | 1 | 20 | C | | | M | 1105 | | | 6 |
| 7 | | TEMP CONTROL PANEL | 200 | E | 1 | 20 | A | | | M | 1105 | | | 8 |
| 9 | | | **1233** | **M** | | | B | 35 | 3 | M | 1105 | HEATING WATER PUMP | | 10 |
| 11 | **1** | **CHILLED WATER PUMP** | **1233** | **M** | 3 | 35 | C | | | M | 1105 | | | 12 |
| 13 | | | **1233** | **M** | | | A | 20 | 1 | | | SPARE | | 14 |
| 15 | | | 4140 | | | | B | 20 | 1 | | | SPARE | | 16 |
| 17 | | PANEL FCU | 5260 | | 3 | 60 | C | 20 | 1 | M | 1176 | BLOWER FAN | | 18 |
| 19 | | | 5120 | | | | A | 20 | 1 | M | 1176 | MAKE UP AIR UNIT | | 20 |
| 21 | | | 396 | M | | | B | | | M | 396 | | | 22 |
| 23 | | HEATING WATER PUMP | 396 | M | 3 | 15 | C | 15 | 3 | M | 396 | HEATING WATER PUMP | | 24 |
| 25 | | | 396 | M | | | A | | | M | 396 | | | 26 |
| 27 | | WATER HEATER | 200 | E | 1 | 20 | B | 20 | 1 | | | SPARE | | 28 |
| 29 | | EXHAUST FAN | 696 | M | 1 | 15 | C | 20 | 1 | | | SPARE | | 30 |
| 31 | | SPARE | | | 1 | 20 | A | 20 | 1 | E | 100 | FIRE/SMOKE DAMPERS | | 32 |
| 33 | | ELECTRICAL/WORK ROOM | 540 | R | 1 | 20 | B | 20 | 1 | R | 180 | ROOF RECEPTACLE | | 34 |
| 35 | | STORAGE/HALL LIGHTS | 750 | L | 1 | 20 | C | 20 | 1 | | | SPARE | | 36 |
| 37 | | WORK ROOM | 360 | R | 1 | 20 | A | | | **M** | **1585** | | | 38 |
| 39 | | WILKINS OFFICE 309 | 540 | R | 1 | 20 | B | 35 | 3 | **M** | **1585** | **CHILLED WATER PUMP** | **1** | 40 |
| 41 | | R-MANAGEMENT HALL 307 | 720 | R | 1 | 20 | C | | | **M** | **1585** | | | 42 |

| LOAD TYPES | CONNECTED LOAD (VA) A | B | C | SUBFEED TOTAL | PANEL TOTAL | DEMAND LOAD (VA) FACTOR | TOTAL |
|---|---|---|---|---|---|---|---|
| LIGHTING | 0 | 0 | 750 | 0 | 750 | 125% | 937.5 |
| RECEPTACLES | 360 | 1260 | 720 | 0 | 2340 | 100% | 2340 |
| MOTORS | 10448 | 9847 | 11579 | 0 | 31874 | 100% | 31874 |
| LARGEST MOTOR | 1233 | 1233 | 1233 | 0 | 3699 | 125% | 4623.75 |
| EQUIPMENT | 300 | 1200 | 528 | 0 | 2028 | 100% | 2028 |
| APPLIANCES | 0 | 0 | 0 | 0 | 0 | 100% | 0 |
| | | | | | **TOTAL DEMAND LOAD (KVA) =** | | **41.8** |
| | | | | | **TOTAL DEMAND LOAD (A) =** | | **116** |

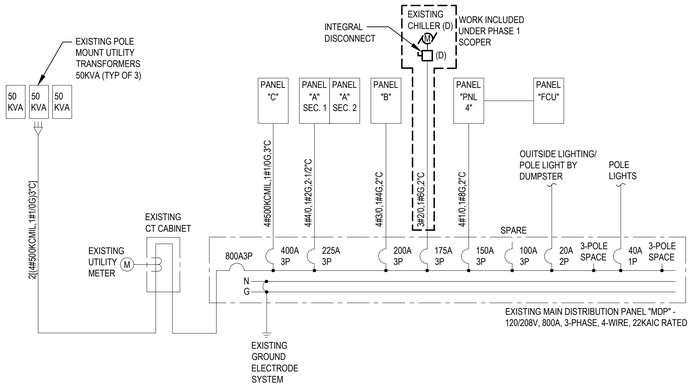
**GENERAL NOTES:**

A. EVERY CIRCUIT AND CIRCUIT MODIFICATION SHALL BE UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.

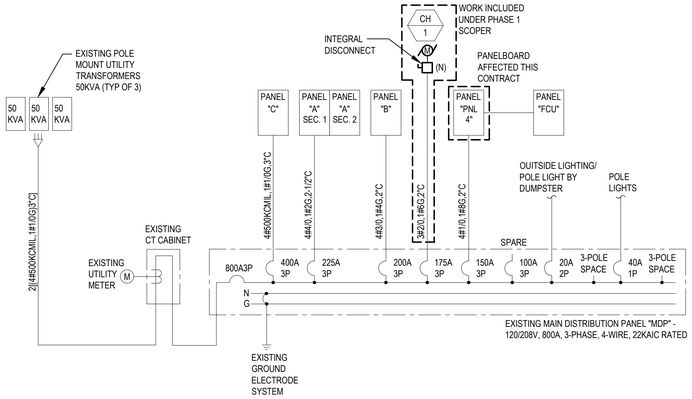
B. PROVIDE NEW UPDATED PANEL DIRECTORIES FOR PANELS THAT ARE INCLUDED IN THIS PROJECT.

**SPECIFIC NOTES:**

1. CIRCUIT AFFECTED THIS CONTRACT BY EQUIPMENT REPLACEMENT. LOAD ADDED IS EQUAL TO OR LESS THAN LOAD REMOVED.




1 E500 ELECTRICAL ONE-LINE DIAGRAM (DEMOLITION)
SCALE: NONE




1 E500 ELECTRICAL ONE-LINE DIAGRAM (NEW WORK)
SCALE: NONE

## PANEL: FCU (EXISTING)

| | |
|---|---|
| **FED FROM:** | PNL 4 |
| **LOCATION:** | SEE PLANS |
| **MOUNTING:** | SURFACE |
| **ENCLOSURE:** | NEMA 1 |

| | |
|---|---|
| **VOLTAGE:** | 208Y/120V, 3PH, 4W |
| **BUS AMPACITY:** | 125A |
| **MAIN:** | MLO |
| **AIC:** | 10,000A |

| CKT NO | NOTES | DESCRIPTION | LOAD VA | LOAD TYPE | BREAKER POLE | BREAKER AMPS | PHASE | BREAKER AMPS | BREAKER POLE | LOAD TYPE | LOAD VA | DESCRIPTION | NOTES | CKT NO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | FAN COIL UNIT 12 | 1080 | M | 1 | 20 | A | 20 | 1 | M | 100 | FAN COIL UNIT 7 | | 2 |
| 3 | | FAN COIL UNIT 13 | 1080 | M | 1 | 20 | B | 20 | 1 | M | 1080 | FAN COOIL UNIT 8 | | 4 |
| 5 | | FAN COIL UNIT 14 | 1080 | M | 1 | 20 | C | 20 | 1 | M | 1080 | FAN COIL UNIT 9 | | 6 |
| 7 | | SPARE | | | 1 | 20 | A | 15 | 1 | M | 800 | FAN COIL UNIT 4 | | 8 |
| 9 | | FAN COIOL UNITS 10/11 | 1220 | M | 1 | 20 | B | 15 | 1 | M | 800 | FAN COIOL UNIT 5 | | 10 |
| 11 | | FAN COIL UNIT 15 | 1080 | M | 1 | 20 | C | 15 | 1 | M | 800 | FAN COIL UNIT 6 | | 12 |
| 13 | | FAN COIL UNIT 16 | 1080 | M | 1 | 20 | A | 20 | 1 | M | 1080 | FAN COOIL UNIT 1 | | 14 |
| 15 | | SPARE | | | 1 | 20 | B | 20 | 1 | M | 1080 | FAN COIL UNIT 2 | | 16 |
| 17 | | SPARE | | | 1 | 20 | C | 20 | 1 | M | 1080 | FAN COIL UNIT 3 | | 18 |

| LOAD TYPES | CONNECTED LOAD (VA) A | B | C | SUBFEED TOTAL | PANEL TOTAL | DEMAND LOAD (VA) FACTOR | TOTAL |
|---|---|---|---|---|---|---|---|
| LIGHTING | 0 | 0 | 0 | 0 | 0 | 125% | 0 |
| RECEPTACLES | 0 | 0 | 0 | 0 | 0 | | 0 |
| MOTORS | 3060 | 4180 | 4040 | 0 | 11280 | 100% | 11280 |
| LARGEST MOTOR | 1080 | 1080 | 1080 | 0 | 3240 | 125% | 4050 |
| EQUIPMENT | 0 | 0 | 0 | 0 | 0 | 100% | 0 |
| APPLIANCES | 0 | 0 | 0 | 0 | 0 | 100% | 0 |
| | | | | | **TOTAL DEMAND LOAD (KVA) =** | | **15.3** |
| | | | | | **TOTAL DEMAND LOAD (A) =** | | **43** |

**GENERAL NOTES:**

A. EVERY CIRCUIT AND CIRCUIT MODIFICATION SHALL BE UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.

B. PROVIDE NEW UPDATED PANEL DIRECTORIES FOR PANELS THAT ARE INCLUDED IN THIS PROJECT.

C. INDICATE SPARES IN PENCIL.

D. ALL NEW CIRCUIT BREAKERS SHALL MATCH EXISTING PANELBOARD MANUFACTURER AND AIC RATINGS.

STAMP

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# CHILLER REPLACEMENT

Jefferson County - Boettcher Mansion
Lookout Mountain, 900 Colorow Rd, Golden, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |

SHEET NAME

ELECTRICAL DETAILS

SHEET NO.

E-500


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